Which constitutive model do measured cavitation records prefer?

Spencer H. Bryngelson

August 7, 2026

Abstract

We compare seventeen constitutive models against laser-induced cavitation records in gelatin at three temperatures, using Bayesian model selection with a noise scale measured from trial-to-trial spread and a prior that prices a model down where it merely reproduces a simpler one it contains. A quadratic Zener solid – strain-stiffening elasticity *and* stress relaxation – wins at every temperature, reaching $\chi^2/N = 1.23$ to 1.35 on the selection grid across the three temperatures, and 0.96 when refitted to its true optimum at 15 °C. Neither ingredient suffices alone: at that temperature relaxation without stiffening reaches 1.98, and both memoryless candidates land at 20 – where they fit each other to one part in a thousand, so the stiffening term does no work at all without a relaxation mode to act on. The advantage of stiffening collapses as the gel warms, which is the clearest physical signal in the comparison.

We report the limitations plainly. The residuals are ceilings rather than estimates, because the winning model's best-fitting region is partly unintegrable. More importantly, a good χ^2/N overstates how well the model does: the residual has lag-one autocorrelation 0.918, so the 201 samples carry the information of about ten independent ones, and what remains is structure the model lacks rather than noise it failed to average away. The ranking survives this (3% change under a correlated likelihood); the absolute claim does not.

We then ask what experiment would do better. Scored by parameter precision and by model discrimination, the criteria disagree, and both are shown to be delicate: a criterion defined by an inner minimum is multimodal here and every improvement to the optimiser lowers it. Reformulating over design *measures* rather than single designs makes the problem convex, and the Kiefer–Wolfowitz equivalence theorem then certifies the answer. Over 480 candidates the optimum splits the runs three ways and beats the best single design by a D-efficiency factor of 5.74 – certified globally optimal at a gap of 9×10^{-10}, not merely the best found.

1 Setup

Each dataset is a set of repeated laser-induced cavitation events at one temperature, recorded as $R(t)/R_{\max}$. Figure 1 shows the mean trace and the trial-to-trial spread, which is what we use as the measurement noise: it is measured rather than assumed, so a residual of $\chi^2/N \approx 1$ means a model reproduces the record to within experimental repeatability.

Two samples are excluded before fitting. A trial reading above its own maximum radius is a tracking failure, and the $t = 0$ sample where every trial agrees exactly is a definition rather than a measurement – giving it an error bar would invent an uncertainty and let it constrain the fit.

Each candidate is evaluated on a log-spaced grid over its own parameters, at one resolution for every model so that grid luck cannot decide the winner. The evidence is a grid quadrature over that box, with the noise scale marginalised under a half-Cauchy prior, a redundancy factor that down-weights parameters where a model merely reproduces a simpler model it contains, and a BIC-style complexity penalty. Trajectories are compressible (Keller–Miksis [1]); the incompressible approximation changes the residuals by a factor of two to four and must not be used for these records.

2 What is being computed, and why

Four objects carry the analysis. Each is derived here rather than quoted, because the choices between them are not neutral and two of the errors this study corrected came from treating them as interchangeable.

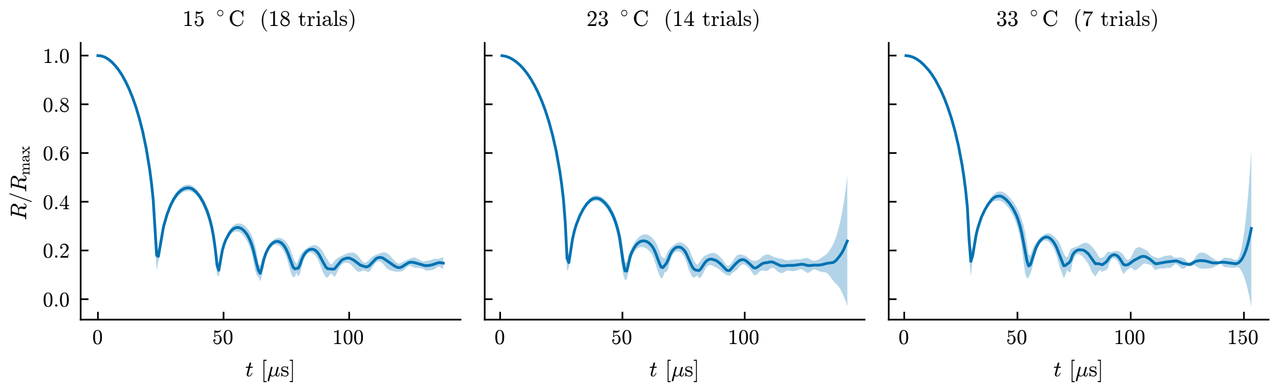

Figure 1: Measured records. Line is the mean over trials, band is ± one standard deviation across trials at each time. The spread is the noise model.

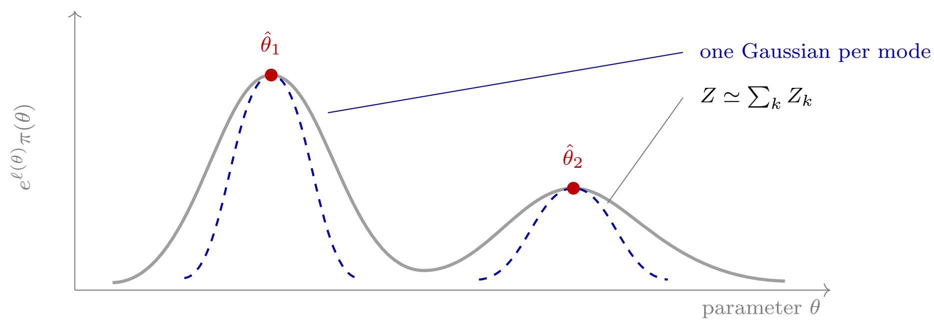

Figure 2: The Laplace expansion, and why the evidence is summed over modes. The integrand $e^{\ell}\pi$ is expanded about each maximum as a Gaussian, whose height and width give that mode's contribution Z_k in closed form; the evidence is $\sum_k Z_k$, computed as a `logsumexp` over the distinct endpoints a multistart finds. Taking only the best mode discards the rest of the integral, which is the difference between an integral criterion and a minimum: a missed mode biases the total slightly, where a missed basin corrupts a minimum entirely. This is why `fit_candidate` returns every distinct endpoint rather than its winner.

2.1 The likelihood, and what it assumes

Each record gives a mean radius y_i at time t_i over 18 trials, with the trial-to-trial standard deviation s_i as the noise estimate. Modelling the observations as independent and Gaussian about the prediction $m_i(\theta)$,

$$p(y \mid \theta) = \prod_i \frac{1}{\sqrt{2\pi}\,\sigma_i} \exp\left[-\frac{(y_i - m_i(\theta))^2}{2\sigma_i^2}\right], \qquad -2\log p = \chi^2(\theta) + \sum_i \log\bigl(2\pi\sigma_i^2\bigr). \tag{1}$$

Three assumptions are buried here and each is separately checkable. *Gaussianity* is the weakest claim, and is reasonable because y_i is a mean over 18 trials. *Known* σ_i is not innocuous – using the sample spread as if it were the true deviation understates uncertainty – and is handled in Section 2.2 by giving the noise scale a prior and integrating it out. *Independence* is the one that fails: the diagonal covariance in (1) says each residual carries fresh information, and Section 9 measures a lag-one autocorrelation of 0.90.

The deviations are not constant across the record. Optical tracking degrades where the wall moves fastest, so the variance is inflated by a logistic weight in the strain rate, $\sigma_i^2 = s_i^2/w_i$ with $w = w_0 + (1-w_0)\,\mathrm{logit}^{-1}\bigl(\kappa(\dot\varepsilon_{\mathrm{th}} - |\dot\varepsilon_i|)/\dot\varepsilon_{\mathrm{th}}\bigr)$. This is a statement about the measurement, not the model, and it is why χ^2/N is dominated by the late record where the weights are large.

2.2 Integrating out the noise scale

Rather than trust s_i, introduce a single scale β multiplying every deviation, $\sigma_i \to \beta\sigma_i$. Substituting into (1),

$$\log p(y \mid \theta, \beta) = -\frac{\chi^2(\theta)}{2\beta^2} - N\log\beta + \text{const}, \tag{2}$$

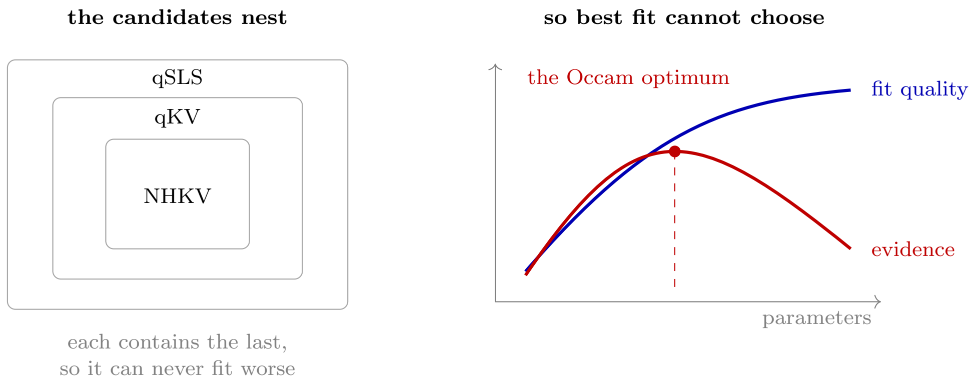

Figure 3: Why this study ranks by evidence and not by fit. The constitutive candidates nest – `NHKV` is `qKV` at zero stiffening and `qSLS` at zero relaxation time – so a larger model can never fit worse, and the best-fitting candidate is simply the most flexible one. The evidence adds the Occam factor of Figure 19, which charges a model for prior volume the data does not use, and so turns a monotone quantity into one with an interior maximum. Everything downstream depends on that penalty being computed correctly, which is why the pathology in Figure 19 matters.

which depends on the data only through χ^2 and N. That is what makes the marginalisation cheap: the forward model never re-enters, so

$$\log p(y \mid \theta) = \log \int_0^\infty p(y \mid \theta, \beta)\, \pi(\beta)\, \mathrm{d}\beta \tag{3}$$

costs one pass over a quadrature grid rather than one solve per node. With the half-Cauchy prior $\pi(\beta) = (2/\pi)/(1+\beta^2)$ the integral is done numerically on $\beta \in [0.05, 10]$.

Why marginalise rather than fit β? Profiling at its best value gives $-\frac{1}{2}N\,[1+\log(\chi^2/N)]$, and marginalising is always smaller. The gap is an Occam penalty: a model that fits only by inflating its own error bars is charged for the inflation. Fitting β would hand that flexibility over for free.

2.3 Why the evidence, and not the best fit

From Bayes' theorem, the posterior odds between two models are the prior odds times the ratio of

$$Z_M = p(y \mid M) = \int p(y \mid \theta, M)\, \pi(\theta \mid M)\, \mathrm{d}\theta. \tag{4}$$

This is the only quantity that answers "which model should I believe" without an arbitrary complexity penalty. Comparing best fits cannot: a nested model can never fit worse than the model containing it, so χ^2 ranks by flexibility. Information criteria add a penalty – $2k$ for AIC, $k \log N$ for BIC – but BIC is an asymptotic approximation to $-2\log Z$ that assumes a regular, well-identified model with N large, and Section 2.5 shows this one is not identified in a direction. The integral itself makes no such assumption.

The penalty appears explicitly on expanding (4) about the maximum. With $\hat\theta$ the best fit, $H = -\nabla^2 \log p(y \mid \theta)$ evaluated there, and $\theta = \hat\theta + \delta$,

$$Z_M \simeq p(y \mid \hat\theta)\, \pi(\hat\theta) \int \exp\left[-\tfrac{1}{2}\delta^{\mathsf{T}} H \delta\right] \mathrm{d}\delta = \underbrace{p(y \mid \hat\theta)}_{\text{best fit}}\; \underbrace{\pi(\hat\theta)\,(2\pi)^{p/2}\,|H|^{-1/2}}_{\text{Occam factor}}. \tag{5}$$

The Gaussian integral is exact for a quadratic $-\log p$, so (5) is the Laplace approximation and its error is governed by the third derivative. Reading the second factor as $\pi(\hat\theta) \times$ (posterior volume) makes its meaning plain: it is the fraction of the prior the posterior actually occupies, so a model needing fine tuning pays. This is the sense in which a good fit and a credible model differ, and it is what Freund and Ewoldt [2] argue should replace goodness-of-fit ranking in rheology.

One consequence matters later and is easy to miss. Because $|H|^{-1/2}$ grows as the posterior broadens, a sloppy direction *raises* the evidence. Model comparison therefore cannot detect non-identifiability – it rewards it – and the two questions must be asked separately.

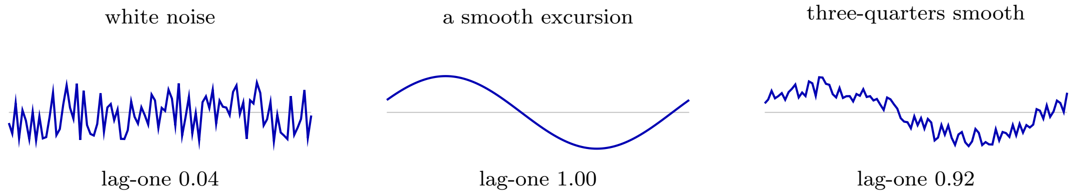

Figure 4: What lag-one autocorrelation measures, and what it does not. Sampled at the $N = 201$ points of these records, lag-one is close to a measure of how much of a residual is *smooth*: white noise gives 0.04, a single smooth excursion 1.00, a mixture that is three-quarters smooth 0.92. It does not say what made the residual smooth – model discrepancy, variation between bubbles and a slow calibration drift are alike in this statistic. The `qSLS` residual gives 0.918; the trial deviations from the mean, with no model fitted at all, give 0.859. Most of the smoothness the fit residual shows is therefore already in the data.

2.4 The prior this study uses

Equation (4) needs $\pi(\theta \mid M)$, and a uniform box would make the answer depend on where the box was drawn. Two factors are used instead, both computed on the same grid the quadrature runs over.

The *bottleneck* maps each parameter to $u_j \in [0, 1]$ in logarithms – the parameters span decades, so a linear map would encode the bounds rather than the physics – and takes the harmonic mean $H(\theta) = p \big/ \sum_j u_j^{-1}$. The harmonic mean is chosen over the arithmetic one precisely because it is dominated by its smallest argument: a parameter driven to the bottom of its range drags the whole prior down, on the reasoning that a parameter the data pushed to a boundary is one the model did not need.

The *redundancy* weight asks whether a model is merely emulating a simpler one it contains. For a candidate and each nested child, the relative stress mismatch F is mapped by $F^n/(F^n + \tau^n)$ against the resolvable scale τ, giving a number near 0 when the parent is imitating the child and approaching 1 when it does something genuinely different. The weight is the *minimum* over children: a model that can imitate any one simpler model is redundant there, however different it looks from the rest.

2.5 The Fisher information, and what its eigenvectors are

Differentiating $-\log p$ twice at fixed data and taking the expectation over noise, the term containing $\partial^2 m$ vanishes because $\mathbb{E}[y_i - m_i] = 0$, leaving

$$F_{jk} = \sum_i \frac{1}{\sigma_i^2} \frac{\partial m_i}{\partial \theta_j} \frac{\partial m_i}{\partial \theta_k} = \left(J^\mathsf{T} J\right)_{jk}, \qquad J_{ij} = \frac{1}{\sigma_i} \frac{\partial m_i}{\partial \theta_j}, \tag{6}$$

so the Fisher information is exactly the Gram matrix of the whitened sensitivities – no approximation, given (1). Near the optimum $\Delta\chi^2 \simeq \Delta\theta^\mathsf{T} F \Delta\theta$, so the surface of constant misfit is the ellipsoid $\Delta\theta^\mathsf{T} F \Delta\theta = c$. Diagonalising $F = V \Lambda V^\mathsf{T}$, its axes are the eigenvectors and their half-lengths are $\sqrt{c/\lambda_k}$: a small eigenvalue is a long axis, a direction along which the data barely object.

The Cramér–Rao bound gives this statistical force. For any unbiased estimator, $\mathrm{Cov}(\hat\theta) \succeq F^{-1}$, so F^{-1} is the best achievable covariance and its eigen-decomposition is the confidence ellipsoid. The standard deviation along direction k is $\lambda_k^{-1/2}$, and the ratio $\kappa = \lambda_{\max}/\lambda_{\min}$ measures how far the problem is from determining all directions equally.

Two conventions are load-bearing. Taking $\theta = \log(\text{parameters})$ makes an eigenvector a power law: moving along v sends $\Delta \log g = v_g t$ and $\Delta \log \alpha = v_\alpha t$, so $g \propto \alpha^{v_g/v_\alpha}$ and the exponent is read directly off the components. And if the sensitivities are taken with respect to a scaled coordinate $u_j = (\log \theta_j - a_j)/w_j$, then $\Delta \log \theta_j = w_j \tilde v_j$ and the exponent is $(w_g \tilde v_g)/(w_\alpha \tilde v_\alpha)$ – with box widths of three and four decades here, a 25% correction, and getting it wrong is what produced a spurious agreement earlier in this work.

Finally, F is the observability Gramian of the system augmented with $\dot\theta = 0$: identifiability and observability are the same statement. The distinction that matters is *structural* unidentifiability, where F is singular and no data can help, against *practical* sloppiness, where F is invertible but ill-conditioned. Here $\lambda_{\min} = 5.2 \times 10^{-2} \neq 0$ and the profile likelihood has a genuine minimum, so the problem is the second – which is why a different experiment can fix it.

2.6 What correlated residuals cost

Equation (1) counts N independent pieces of information. If instead the residuals have autocorrelation ρ_k at lag k, then for a sum of N correlated terms $\mathrm{Var}(\sum r_i) = N\sigma^2\big(1 + 2\sum_k (1 - k/N)\rho_k\big)$, so the information is that of

$$N_{\text{eff}} = \frac{N}{1 + 2\sum_{k\geq 1} \rho_k} \tag{7}$$

independent samples, and parameter uncertainties are understated by $\sqrt{N/N_{\text{eff}}}$. The sum is truncated at the first negative ρ_k – Geyer's initial positive sequence – because the long tail is estimated from few pairs each and contributes sampling error rather than signal.

The correct remedy is a covariance with off-diagonal terms, not a larger σ. Taking $C_{ij} = \sigma_i\sigma_j \exp(-|t_i - t_j|/\tau)$, the Cholesky factor $C = LL^\mathsf{T}$ gives the generalised least squares form

$$-2\log p = r^\mathsf{T} C^{-1} r + \log|C| + \text{const} = \|L^{-1}r\|^2 + 2\sum_k \log L_{kk} + \text{const}, \tag{8}$$

which is what `FieldObservation` computes. Inflating σ instead would treat structure as noise: it widens the error bars while leaving the parameter estimates biased, which is the failure Brynjarsdóttir and O'Hagan warn about.

2.7 Which design criterion, and why they disagree

An experiment is chosen by a scalar summary of F, and the classical family is

$$\text{D: } \max\det F, \qquad \text{A: } \min\operatorname{tr} F^{-1}, \qquad \text{E: } \max\lambda_{\min}, \qquad \text{c: } \min c^\mathsf{T} F^{-1} c. \tag{9}$$

D minimises the volume of the confidence ellipsoid, A its mean squared axis length, E its longest axis, and c the variance of one nominated combination. They rank designs differently because they summarise the same spectrum differently, and the choice is a statement about what one wants to learn.

For identifiability the relevant fact is a scaling argument. A design change that only amplifies signal – more samples, less noise, a larger bubble – sends $F \to sF$. Then $\det F \to s^p \det F$ and $\lambda_{\min} \to s\lambda_{\min}$, so D and E both improve, while

$$\kappa = \frac{\lambda_{\max}}{\lambda_{\min}} \quad \text{is unchanged.} \tag{10}$$

Degeneracy is a property of the eigen*vectors*, not the eigenvalues, and only a design that rotates them can remove it. This is exactly why the E-optimal search preferred a larger bubble while conditioning preferred a gentler collapse: more signal against better separation.

The c-optimal choice carries a smallr trap. Nominating c as the sloppy direction *of the present design* optimises for a direction that moves once the design changes, and in this study that criterion recommended the opposite of the right answer.

When the model itself is in doubt, none of these apply, since all presuppose the model whose F is being computed. T-optimality asks instead for the design maximising the discrepancy between rival predictions, $\max_\xi \|m_1(\hat\theta_1;\xi) - m_2(\hat\theta_2;\xi)\|$, and privileges neither.

3 What the records determine

The winner fits, but not every parameter it contains is measured by these data.

Why the trade exists

Write $\hat g = G/p_\infty = 1/\mathrm{Ca}$ for the nondimensional modulus and $\lambda = R_{\text{eq}}/R$ for the stretch the stress law sees. The equilibrium stress the Maxwell arm relaxes toward is

$$Z_e = R^3\left[\tfrac{3\alpha-1}{2}\,\hat g\, P(\lambda) + 2\alpha\,\hat g\, Q(\lambda)\right], \qquad \begin{aligned} P(\lambda) &= 5 - \lambda^4 - 4\lambda, \\ Q(\lambda) &= \tfrac{27}{40} + \tfrac{1}{8}\lambda^8 + \tfrac{1}{5}\lambda^5 + \lambda^2 - \tfrac{2}{\lambda}. \end{aligned} \tag{11}$$

Collecting powers of α separates it into two terms with different parameter dependence,

$$Z_e = \hat{g}\,\alpha\,A(\lambda) + \hat{g}\,B(\lambda), \qquad A = R^3\left[\tfrac{3}{2}P + 2Q\right], \qquad B = -\tfrac{1}{2}R^3 P. \tag{12}$$

This is exact; it reproduces the implemented expression to 2.5×10^{-14} over random states. Only the first term carries α, so $\hat{g}\alpha$ and $\hat{g}$ are separately determined only while *both* terms contribute.

They do not, at depth. As λ grows, $Q \to \frac{1}{8}\lambda^8$ dominates every other term in either bracket, so $A \to \frac{1}{4}R^3\lambda^8$ while $B \sim \frac{1}{2}R^3\lambda^4$, and

$$Z_e \;\longrightarrow\; \tfrac{1}{4}\,\hat{g}\,\alpha\,R^3\lambda^8, \tag{13}$$

a function of the *product* $\hat{g}\alpha$ alone. The likelihood cannot then distinguish a soft material that stiffens sharply from a stiff one that barely stiffens. Holding Z_e fixed in (12) gives the trade in closed form,

$$\frac{\partial \log \hat{g}}{\partial \log \alpha} = -\frac{\alpha A}{\alpha A + B} \;\longrightarrow\; -1 \quad (\lambda \to \infty), \tag{14}$$

and the share of the signal that separates the two parameters is

$$\frac{|\hat{g}B|}{|\hat{g}\alpha A|} = \frac{|B|}{\alpha |A|} \simeq \frac{2}{\alpha\lambda^4}. \tag{15}$$

Equations (14) and (15) are what the measurements see. These records collapse to $\lambda = 2.215$, where (14) gives -0.981 against -0.99 from the Fisher curvature – a prediction from the constitutive law, not a fit. The profile likelihood returns -0.80 because it averages over a range of λ, and the exponent is -0.861 at $\lambda = 1.5$. Separability falls as λ^{-4}: it is 2% of the signal at $\lambda = 2.2$ and 16% at $\lambda = 1.5$.

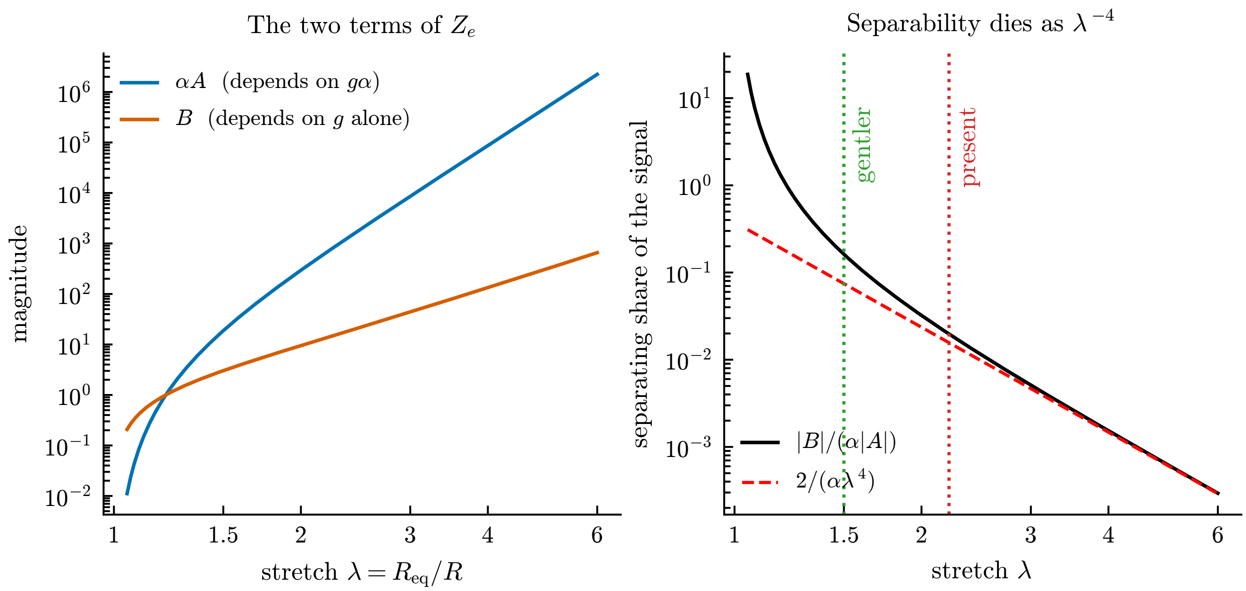

Figure 5: The two terms of (12) against collapse depth. Left: αA, which depends on the product $g\alpha$, outruns B, which carries g alone. Right: their ratio, the share of the signal that can separate the two parameters, following $2/(\alpha\lambda^4)$ once the collapse is deep. It is 2% at the depth these records reach and 16% at $\lambda = 1.5$.

What the Fisher information measures

For Gaussian observations with known deviations the log-likelihood is $-\frac{1}{2}\sum_i (y_i - m_i(\theta))^2/\sigma_i^2$ up to a constant, so with the whitened sensitivity $J_{ij} = \sigma_i^{-1}\partial m_i/\partial\theta_j$ the Fisher information is exactly $F = J^{\mathsf{T}}J$, and $\Delta\chi^2 \simeq \Delta\theta^{\mathsf{T}}F\Delta\theta$. Taking θ to be the logarithms of the parameters makes the eigenvectors read as power-law combinations: along an eigenvector v the ratio v_g/v_α is the exponent in $g \propto \alpha^{v_g/v_\alpha}$. This is also the observability Gramian of the system augmented with $\dot{\theta} = 0$, which is why identifiability and observability are the same statement here.

One caution, since it caught us. When the sensitivities are taken with respect to a unit cube $u_j = (\log\theta_j - a_j)/w_j$ rather than $\log\theta_j$ directly, the eigenvector components carry the box widths: $\Delta\log\theta_j = w_j\,\tilde{v}_j$, so the exponent is $(w_g\tilde{v}_g)/(w_\alpha\tilde{v}_\alpha)$, not $\tilde{v}_g/\tilde{v}_\alpha$. With w of three and four decades the two differ by 25%.

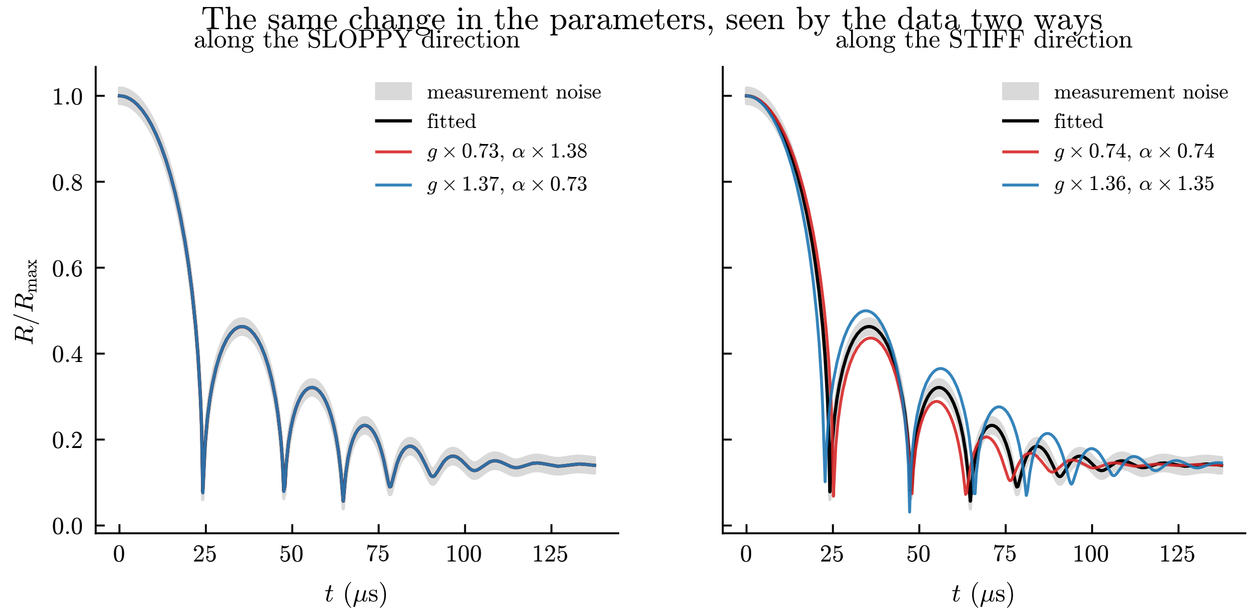

Figure 6: What an eigenvector means, shown rather than defined. The parameters are moved the same distance in log-space along each eigenvector. Along the sloppy direction a 27% change in the modulus, compensated by 38% in the stiffening, leaves the trajectory inside the measurement noise; along the stiff direction a change of the same size is obvious. The data cannot see the first and cannot miss the second.

Modulus and stiffening trade against each other. Two independent analyses agree. The Fisher information at the fitted optimum has an eigenvalue spectrum spanning six decades, and the eigenvector of its smallest eigenvalue is almost purely a shear-modulus-against-stiffening direction, with weights -0.796 and $+0.604$ and contributions of -0.003 and -0.032 from viscosity and relaxation time. Separately, profiling the likelihood in α – fixing it and re-optimising everything else – gives $g \propto \alpha^{-0.80}$ across two decades. Sampling the posterior gives a third, independent measurement: its widest direction is $(-0.682, +0.002, -0.082, +0.727)$ in the logarithms of $(g, \mu, \lambda_1, \alpha)$, and $\log g$ is 98.7% anticorrelated with $\log\alpha$.

The three exponents are -0.99 from local curvature, -0.94 from the posterior covariance, and -0.80 from the profile. They agree that the trade is near-reciprocal and disagree in the third digit, which is expected rather than troubling: the valley is curved, so a curvature estimate at one point and a chord measured across two decades are not measuring the same slope. The correlation is the more robust statement.

So viscosity and relaxation time are cleanly determined, while modulus and stiffening are constrained only in combination. The identifiable quantity is closer to $g\,\alpha^{0.8}$ than to either alone, and a fitted α should be reported with its correlation to g rather than as an independent material property.

Figure 8 shows why the four estimates in this study disagree about g by an order of magnitude while agreeing about the fit: they are different points on the floor of a single flat-bottomed trench.

The stiffening is nonetheless identified, once the prior allows it. Sampling with the ceiling on α placed at 1, 10 and 100:

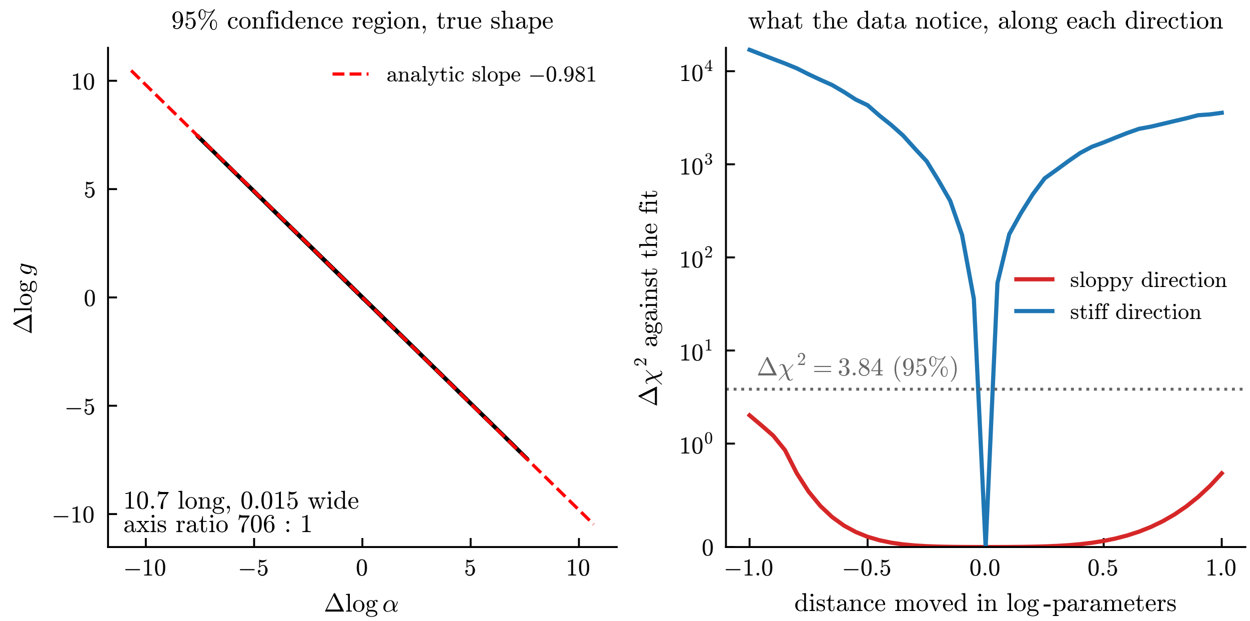

Figure 7: Left: the 95% region from the inverse Fisher information, 10.7 long and 0.015 wide in log-parameters – an axis ratio of 706, which is why it draws as a line, and the analytic slope of -0.981 lies along it. Right: the misfit measured by re-solving at perturbed parameters, not the quadratic approximation. Moving along the stiff direction costs 10^4 in $\Delta\chi^2$; moving the same distance along the sloppy one stays below the 95% threshold, so the parameters may shift by a factor of $e \approx 2.7$ without the data objecting.

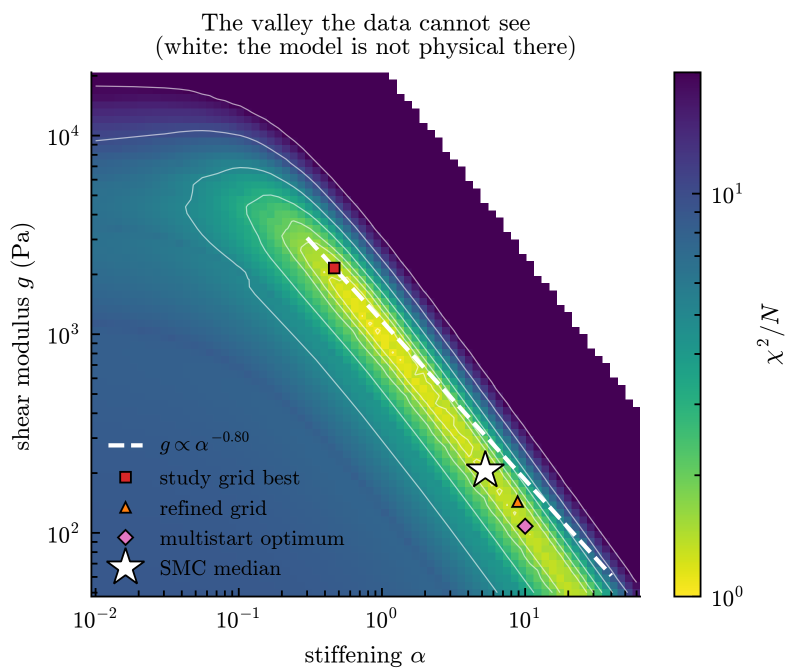

Figure 8: χ^2/N over the modulus–stiffening plane, at the fitted viscosity and relaxation time. The dashed line is $g \propto \alpha^{-0.80}$, predicted from the Fisher curvature at a single point; it lies along the floor of the valley. Four "best fits" from four methods are strung out along that floor rather than disagreeing with one another. The white region at upper right is where the model is not physical.

ceiling	log evidence	median α	97.5%
1	2162.9	0.85	0.97
10	2173.9	6.37	9.04
100	2172.4	5.27	8.73

Widening the prior tenfold moves the median only from 6.37 to 5.27 and the evidence by 1.5 out of 2173, so α is identified near 5 with 97.5% of posterior mass below about 9. A prior-dominated parameter would have tracked its bound. The ceiling of 1 shows what genuine truncation looks like: the median pins against it and 11 log units of evidence are lost.

This corrects a natural reading of the profile likelihood alone, which appears flat to the boundary only because the boundary sits where the posterior's upper tail does.

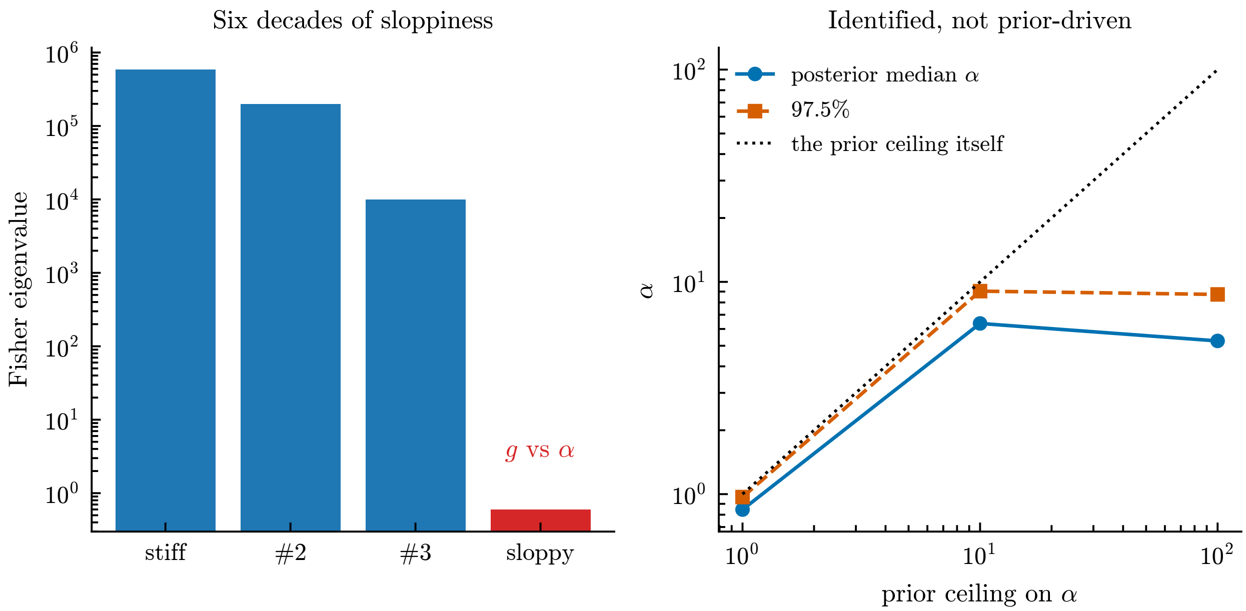

Figure 9: Left: the Fisher spectrum at the optimum, six decades between the stiffest and sloppiest directions, with the sloppiest being modulus against stiffening. Right: the posterior for α against the ceiling imposed on its prior. It tracks the ceiling while truncated and departs from it once the ceiling is loose, which is what distinguishes an identified parameter from a prior-driven one.

4 Result

Table 1 and Figure 10 give the outcome. The quadratic Zener model `qSLS` – strain-stiffening elasticity with stress relaxation – wins at every temperature, and the ordering beneath it separates cleanly into three tiers: models with both ingredients, models with relaxation only, and memoryless models.

Table 1: Best χ^2/N *on the selection grid* by model class – the best node, not the best fit; Section 4.2 refits and reaches 0.962 for qSLS at 15 °C. Values near 1 indicate a fit within experimental repeatability.

	15 °C	23 °C	33 °C
`qSLS` (stiffening + relaxation)	**1.35**	**1.33**	**1.23**
`SLS` (relaxation, linear elasticity)	2.94	1.85	1.41
viscoelastic fluids	5.3–5.7	4.5–4.7	1.9–2.1
memoryless	12.6–16.7	–	–

Figure 11 overlays the winning model on the data it was selected against.

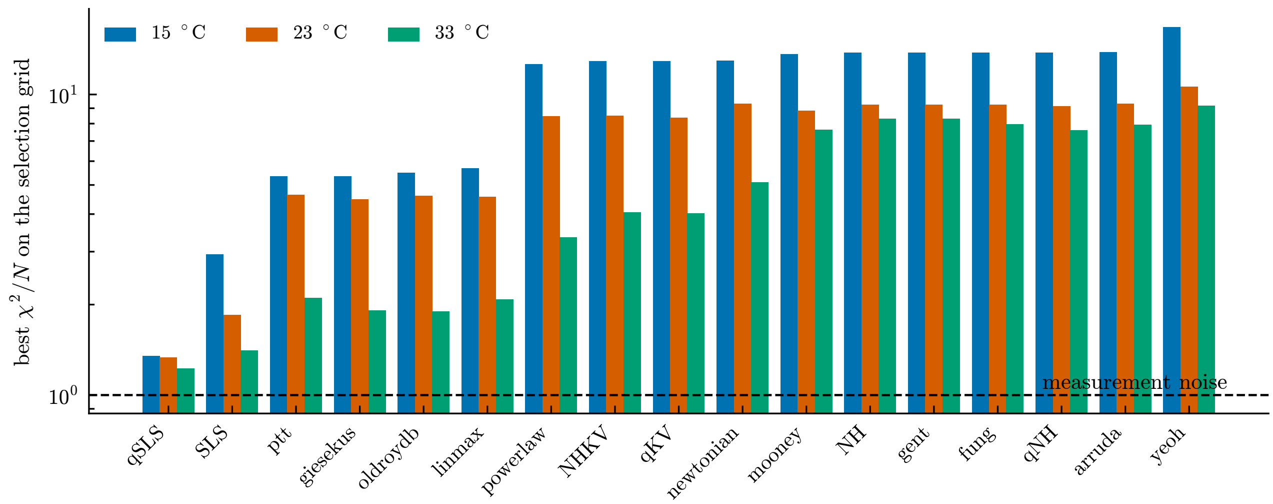

Figure 10: Best χ^2/N on the selection grid for every candidate, all three temperatures, logarithmic scale. The dashed line is the measurement-noise floor.

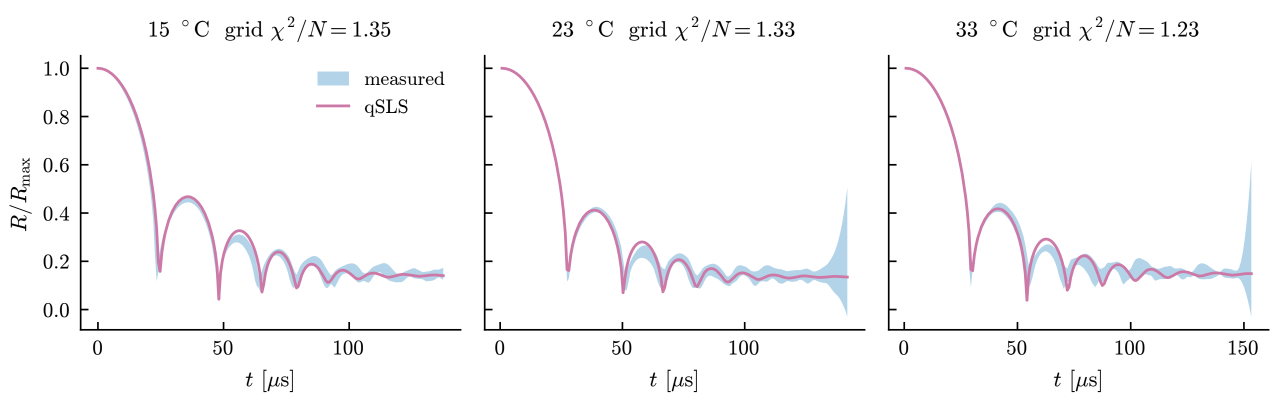

Figure 11: The selected model against the measured band it was fitted to.

4.1 Reading the posteriors

The model posteriors saturate: the winner takes essentially all the mass and the runners-up are reported as 10^{-29} or smaller. These magnitudes should not be read literally. Evidence differences scale with sample count, and with roughly 3600 samples a modest per-sample difference in fit compounds into an astronomical Bayes factor. The defensible reading is ordinal – the winner is decisive and the ordering is meaningful, the magnitudes are not. The residual χ^2/N is the quantity to quote.

4.2 How good are the fits, really?

The table above reports the best χ^2/N found on the evaluation grid. Two separate questions follow, and they have different answers: how well does the winning model fit, and how much does a good χ^2/N entitle us to claim?

The grid understates the fit. The model-selection grid evaluates each candidate at log-spaced parameter nodes, so its best value is the best *node*, not the best fit. Refitting each model to the 15 °C record by multistart Gauss–Newton with analytic Jacobians – eight starts from a Latin hypercube over the wide prior box, taking the lowest – gives:

model	parameters	χ^2/N	lag-one ρ	N_{eff}	χ^2/N_{eff}	inflation
qSLS	4	**0.962**	0.918	10.3	18.9	4.43
SLS	3	1.984	0.932	7.1	56.6	5.34
qKV	3	20.18	0.948	13.4	302	3.87
NHKV	2	20.16	0.948	13.4	302	3.87

At its true optimum qSLS reaches $\chi^2/N = 0.96$ rather than the grid's 1.35, which confirms in a second way that the tabulated residuals are ceilings. The ordering is unchanged, and the gaps are large: relaxation without stiffening doubles the residual, and both memoryless models sit twenty times above it.

Stiffening buys nothing without relaxation. qKV and NHKV differ by the stiffening exponent α and by nothing else, and they fit identically – 20.18 against 20.16, a difference of one part in a thousand on a residual twenty times too large. The strain-stiffening term, which is decisive in the presence of a relaxation mode, is worth nothing without one. That is a sharper statement than the evidence ranking makes on its own: it is not that qKV is a worse model than qSLS, but that its extra parameter does no work at all. It also explains why the temperature trend is carried by the interaction of the two mechanisms rather than by either alone.

But $\chi^2/N \approx 1$ does not mean the fit is within noise. The statistic χ^2/N compares the residual to its expectation under the assumption that the N residuals are independent draws of known scale. They are not. The lag-one autocorrelation of the qSLS residual is 0.918, and by the Geyer initial-positive-sequence estimator [3] of (7) the 201 samples carry the information of about $N_{\text{eff}} = 10$ independent ones. Dividing by that count instead gives 18.9.

That number should not be read as a calibrated test – for correlated residuals the correct statistic is $r^\mathsf{T}\Sigma^{-1}r$ with the true covariance, not χ^2 rescaled by an effective count – but its message is robust and it is the opposite of the headline's. A residual with $\rho = 0.918$ is not noise the model failed to average away; it is *structure* the model does not contain. The fit is excellent in the sense that no candidate does better, and it is not excellent in the sense of being consistent with the measurement error.

Three things follow, and they should be read together.

1. *The ranking survives.* Refitting under a correlated likelihood with a Cholesky whitening of an estimated AR(1) covariance changes the qSLS–SLS evidence gap by 3%. What the correlation destroys is the absolute claim, not the comparison, and the comparison is what the study is for.

2. *The denominator is the wrong yardstick.* The scale σ_i is the trial-to-trial spread, which is substantially variation in the parameters between bubbles rather than error in measuring any one of them. A χ^2 formed against it is therefore not comparing the model to measurement precision at all.

3. *The misfit is in the early record.* It is not, as one might expect, concentrated where the tracking is hardest. Restricting the fit to a later window makes χ^2/N worse, from 0.97 to 1.95, and does not whiten the residual – so the structure cannot be excised by discarding the awkward part of the record.

The honest summary is that qSLS is the best available description of these records by a wide margin, that it is measurably incomplete, and that the size of what is missing is better indicated by the residual's correlation than by its magnitude.

4.3 What the trial spread actually is

The likelihood of Section 2.2 treats the residuals as independent draws whose scale is the trial-to-trial spread s_i. Two assumptions are buried in that, and both are wrong in ways that matter for how the numbers above should be read.

Part of the spread is the parameters, not the measurement. If the 18 trials differed only by measurement error, their deviations from the trial mean would be white and would point nowhere in particular. They do neither. The singular spectrum of the 18×201 deviation matrix puts 53.9% of the variance in one component where white noise would give 5.9%, and projecting each deviation onto the span of $G = \partial(R/R_{\max})/\partial\theta$ for the six parameters $(R_0, R_{\rm eq}, g, \mu, \lambda_1, \alpha)$ captures 39.3% of the total variance against a chance level of $6/201 = 3.0\%$. Noise does not align with a six-dimensional subspace of a 201-dimensional space by accident.

The physical reading is that each bubble has its own parameters, drawn from a population: $\theta_j \sim \mathcal{N}(\theta_{\rm pop}, \Sigma)$. The mean trace then has covariance

$$C \;=\; \frac{1}{J}\Big(G\,\Sigma\,G^{\mathsf{T}} \;+\; \sigma^2_{\rm meas} I\Big), \tag{16}$$

low rank plus diagonal rather than diagonal. Note that Σ itself is not estimable here – regressing the deviations onto G inherits the g–α degeneracy of Section 2.5 and returns $\mathrm{sd}(\log g) = 159$ – but only the prediction covariance $G\Sigma G^{\mathsf{T}}$ enters (16), and that is the sample covariance of the projected deviations, which needs no inversion.

It is not what makes the residuals correlated. The natural hypothesis is that the lag-one correlation of the fit residual is this same effect. Whitening by (16) tests it directly:

covariance	χ^2/N	lag-one ρ
diagonal (independent)	17.54	0.758
hierarchical, (16)	21.96	**0.516**

The correlation falls by about a third and most of it survives; χ^2 gets *worse*, so the residual sits in directions where C is small. The dominant term is therefore model-form error, not variation between bubbles. Both effects are real and the model-error one is larger – which separates two things that Section 9 had conflated.

Which precision χ^2/N is measured against. The same computation exposes a factor of $J = 18$ in the meaning of the headline number. The data is the *mean* of 18 trials, but s_i is the spread *between* trials, and the standard error of the mean is smaller by $\sqrt{18} = 4.24$. From the identical residual,

$$\chi^2/N = 0.974 \;\text{ against the trial spread,} \qquad \chi^2/N = 17.54 \;\text{ against the standard error of the mean.}$$

Both are meaningful, and they answer different questions. *The model reproduces the record to within the scatter between individual experiments* is true, and is the claim this document makes. *The model is statistically adequate* is false: against the precision the 18-trial mean actually carries, it is off by a factor of seventeen. The noise-scale marginalisation of Section 2.2 absorbs the difference – the implied $\beta \approx 0.24$ lies inside its prior – so the evidence ranking is undisturbed. What changes is what a residual near one entitles one to say.

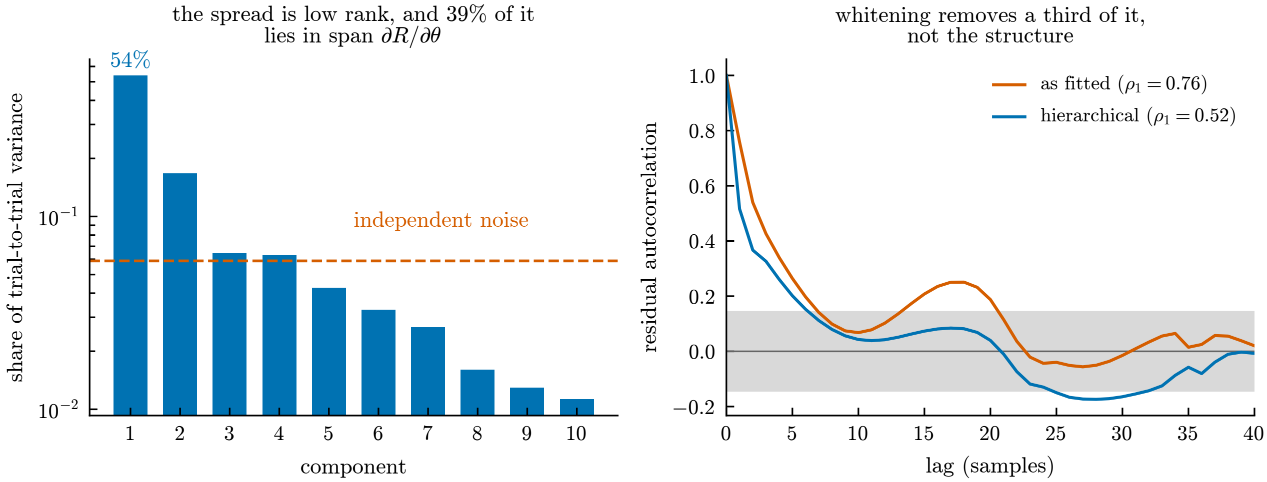

Figure 12: What the trial spread is, and what it does not explain. Left: the singular spectrum of the eighteen trial deviations. Independent measurement error would put an equal share in every component (dashed); instead the leading one carries 54%, and 39% of the total lies in the six-dimensional span of $\partial R/\partial\theta$ against a 3% chance level. Right: the autocorrelation of the qSLS fit residual before and after whitening by the hierarchical covariance (16), with the white-noise band shaded. If the correlation were caused by variation between bubbles, whitening would flatten it into that band. It removes about a third and leaves the rest, so the remainder is structure the model does not contain.

What lag-one does and does not measure. The inference above – that whitening removes a third of the correlation and the remainder is structure the model lacks – rules out variation between bubbles, and nothing more. Lag-one autocorrelation at $N = 201$ samples is close to a measure of how much of a residual is *smooth*, not of what made it smooth: sampled at this density, white noise gives 0.04, a single sine period gives 1.00, a smooth random curve gives 0.999, and a mixture that is 80% smooth gives 0.944. A value of 0.918 says the residual is roughly three-quarters smooth structure. It does not say whose.

That matters because the trial deviations from the mean – pure measurement variation, with no model fitted at all – carry lag-one 0.859 (median over the eighteen trials, range 0.725 to 0.890). Most of the smoothness the fit residual shows is already present in the data before any model is involved. Whitening by the hierarchical covariance removes the part of it that lies along $\partial R/\partial\theta$, which is why it removes only a third; the rest is smooth in the data for reasons the parameter sensitivities cannot represent, and model-form error is one candidate among several rather than the established remainder.

The searches of Section 4.4 were run on the stronger reading, and their negative results are consistent with the weaker one.

4.4 What to enrich, and how it would have to be judged

The previous section leaves a specific defect: the residual carries structure the model does not contain, and it is not accounted for by variation between bubbles. The natural suspect is the assumption of a *single* relaxation time, which is strong for a crosslinked biopolymer whose relaxation is in reality distributed. Two enrichments follow from that suspicion, and they are not the same claim.

A second timescale. `TwoModeQuadraticZener` adds a second Maxwell arm sharing the elastic equilibrium, with a weight w splitting both the elastic target and the viscous forcing,

$$\dot{Z}_1 = -\frac{Z_1 - (1-w)Z_\mathrm{e}}{\mathrm{De}_1} + \frac{(1-w)\,\Phi}{\mathrm{De}_1}, \qquad \dot{Z}_2 = -\frac{Z_2 - wZ_\mathrm{e}}{\mathrm{De}_2} + \frac{w\,\Phi}{\mathrm{De}_2}, \tag{17}$$

where Φ is the forcing the one-mode law already carries. Splitting *both* terms is what makes $w = 0$ exact: the second memory is then driven by nothing, decays from zero, and stays there.

A timescale that moves. `CarreauZener` keeps one arm and lets its relaxation time depend on the shear rate. A Maxwell arm is a spring and a dashpot in series, so $\lambda = \eta/G$; if the dashpot obeys Carreau rather than Newton,

$$\mathrm{De}_{\mathrm{eff}} = \mathrm{De}\left[1 + (\Lambda\dot{\gamma})^2\right]^{(n-1)/2}, \tag{18}$$

and the arm relaxes faster the harder the medium is sheared. A collapse spans decades of shear rate, so this is a genuinely different claim from (17) about the same residual. Both terms of the memory equation carry $\mathrm{De}_{\mathrm{eff}}$, since an arm has one time constant; at $n = 1$ the bracket is unity at every shear rate and the law is identically the one-mode model.

Why the grid cannot judge either. Both carry six free parameters, and the quadrature used throughout is a Cartesian product costing count^p. At the 12 per axis used throughout, six axes is 5 971 968 solves against 168 072 for the entire standard set. The remedy is to expand about a fit rather than quadrature over a grid, at $1 + 2p$ solves – but doing so exposes a defect in the expansion itself.

The Occam factor must be capped at the prior. Written as in (5), the Laplace evidence contains $-\frac{1}{2}\log\det J^{\mathsf{T}}J$. A direction the data does not determine sends an eigenvalue of $J^{\mathsf{T}}J$ to zero and that term to $+\infty$: the expansion *rewards* a model for a parameter its design cannot see. This is not a small correction. On synthetic data generated from the one-mode model, the two-mode model scores

$$+29.7 \text{ nats above the one-mode model at } w = 10^{-3}, \qquad -2.6 \text{ at } w = 0.3,$$

so it wins precisely where its second arm does nothing and loses only where the arm does real work. The cause is the fitted Gaussian extending far outside the bounded cube the prior occupies. Capping each eigendirection's contribution at unity – a posterior cannot be wider than the prior it came from, Section A.6 – returns -0.0000, -0.0004, -0.033 and -6.81 at $w = 10^{-3}, 10^{-2}, 0.05, 0.3$: monotone, and never paying for what the data cannot resolve. The correction is a strict generalisation, agreeing with the plain form to round-off wherever every direction is already sharper than the prior.

The answer, on the $15\,^{\circ}\mathrm{C}$ **record.** Both were fitted by multistart in the prior's unit coordinates, scored by the capped expansion, and summed over the modes the multistart found.

model	p	χ^2/N	$\log Z$	vs. qSLS	lag-one ρ	N_{eff}
qSLS	4	0.962	529.26	–	0.918	10.3
qSLS2	6	0.950	531.03	+1.78	0.918	7.2
qSLSthin	6	0.922	532.05	+2.79	0.875	8.2

The one-mode χ^2/N of 0.962 reproduces Section 4.2 exactly, by a different optimiser in different coordinates, which is the check that the fits are comparable at all.

Both enrichments are preferred, and neither margin means what it appears to. The residual autocorrelation is the defect they were reached for, and it does not move: 0.918 unchanged for the second relaxation time, $0.918 \to 0.875$ for the rate-dependent one, against N_{eff} near 8 out of 201 either way. Both $\log Z$ and χ^2/N presume independent residuals, so the likelihood overstates its own information by roughly $N/N_{\mathrm{eff}} \approx 25$; deflating a 2.79-nat margin by anything of that order leaves nothing. The second arm is not idle – it lands at 16% of the memory with a timescale 15.7 times the first – it simply does not address what is wrong. And `qSLSthin` pins its crossover at the bottom of its range, so that margin is bound-limited and the crossover is not resolved by this record at all.

What this says. A richer relaxation spectrum is not the missing physics, or at least not the part that matters here. Two independent enrichments, each reducing exactly to the one-mode law and each free to switch itself off, both decline to and both leave the correlation in place. That is evidence about where to look next – the residual is concentrated in the first collapse, where sphericity and the far-field boundary are least secure – and it is more useful than a small positive margin would have been.

5 The forward operator

Every comparison above varies the material and holds the bubble-dynamics equation at Keller–Miksis. That is an assumption. With the material at its fit, changing only the operator moves the trace by four to fourteen times the median noise, which exceeds any constitutive difference measured here.

The operators are not candidate materials, so they are compared differently. We hold the candidate fixed and vary the forward solve. The parameter space is then identical across the set, the Occam terms cancel, and the difference in $\log Z$ is a Bayes factor between operators.

The prior box sets the answer. Scored with the bounds of Section 2.4, Keller–Miksis/Mie–Grüneisen ranks first at +4.02 nats and Gilmore/Tait second to last at −27.66. Three of the six fits sit on $g = 10^2$ or $\lambda_1 = 10^{-7}$, their own floors. Widening the box to $g \in (10^0, 10^6)$ and $\lambda_1 \in (10^{-9}, 10^{-2})$ reverses the ranking: Gilmore/Tait leads at +21.93 and KM/Mie–G falls to +7.32.

Widening moves the pinning rather than removing it. Five of six fits then sit on $\alpha = 10^3$ with g collapsed to 1.1 Pa to 3.5 Pa. Figure 20 shows the mechanism: the optimiser slides along the $g\alpha$ valley until it meets an edge.

Identified coordinates. Fixing g/α and fitting the product $g\alpha$ removes the pinning entirely, with none of 54 fits on a bound. We repeat at three ratios spanning two decades and on all three records. An ordering counts as settled when one record decides it by more than a nat and none contradicts it.

ordering	15 °C	23 °C	33 °C
compressible > Rayleigh–Plesset	+162 to +182	+74 to +79	+74 to +75
Gilmore/Tait > Keller–Miksis	**+17.1**	**+4.2**	**+1.2**
Keller–Miksis/Tait > Keller–Miksis	+5.8	+2.0	
Keller–Miksis/Mie–G > Keller–Miksis	+4.5	+1.5	

Twelve of the fifteen pairwise orderings are settled. Compressibility is required on every record. The pressure form assumed by every candidate in this package is beaten by three separate operators, and by Gilmore/Tait on all three records.

A search budget produced an earlier answer. At six multistart restarts the same comparison placed Gilmore/Tait fourth on two records and first on the third, and reported the operator as undetermined. It also settled no ordering at 33 °C, which we read as a record without discriminating power. At ten starts the 33 °C fits reach $\chi^2/N = 0.42$ where six starts found 1.151.

Every operator had been held in the same poor basin. A record that cannot decide and a search that has not converged produce the same output. Only a better search separates them.

6 Three model axes

With the constitutive law, the operator and the thermal treatment all uncertain, the design question is not which experiment discriminates but whether the axes are separable at all. Two ways they need not be. A difference lying in the span of $\partial R/\partial\theta$ is reproduced by refitting the material and is invisible at every design (Figure 15); and two axes whose remainders are nearly parallel move the evidence together, so no design can say which moved it.

axis	size	after refit	absorbed
dynamics (KM → KM/Mie–G)	8.89	**1.98**	**77.7%**
constitutive (one → two modes)	14.42	8.14	43.5%
thermal (cold → bubble + medium)	**15.99**	8.46	47.1%

in units of the record's noise. The remainders sit at 46, 71 and 38° to one another, so no pair is confounded and the design problem is well posed for all three. Two things follow that were not expected. Thermal is the *largest* lever, not a negligible one – it moves the trace most and survives refitting most, so the earlier observation that switching it on made χ^2/N slightly worse says the direction it moves in is not the one the record wants, not that it is small. And the operator is the most absorbed axis by a wide margin, which is the mechanism behind Section 5: most of an operator change can be mimicked by adjusting the material.

A design against all three. Making each axis a parameter through $R(\theta, \varepsilon) = (1-\varepsilon)R_A + \varepsilon R_B$, so that $\partial R/\partial\varepsilon = R_B - R_A$ is one more column of J, puts discrimination and estimation in one information matrix and leaves the criterion concave – the certificate of Figure 13 applies unchanged. The information is averaged over a decade of $g\alpha$, because operator information is governed by collapse depth and a measure certified at one material misranks others. The measure certifies at a gap of 10^{-9} on four support points.

axis	variance at the optimum	at the 15 °C geometry	efficiency
dynamics	6.78×10^{-3} (binding)	9.91×10^{-3}	68.4%
constitutive	5.45×10^{-4}	2.24×10^{-2}	2.4%
thermal	4.40×10^{-4}	1.05×10^{-2}	4.2%

Reported per axis rather than as one determinant, because a design maximising det could buy its whole score on the two easy axes and leave the operator undetermined without saying so. The experiments performed are near-best for the hardest axis and 24 to 41 times off for the two easiest, so a change of geometry buys a great deal of constitutive and thermal discrimination at little cost to the operator: the support wants $R_{\max}$ near 60 µm at high stretch and near 1200 µm at low stretch, against the 277 µm and stretch 7.09 performed.

7 Designing a better experiment

Designing without assuming the model

Everything above optimises the Fisher information of one model at its fitted parameters, which is circular if that model is wrong. Two checks that do not privilege it.

The record is information-rich; the parameterisation is not. Drawing 400 trajectories from the posterior and taking the singular values of the centred ensemble, eleven modes rise above the trial noise floor of $0.018\,R_{\max}$. That is not a count of identifiable combinations – four varied parameters span at most a four dimensional manifold, and the extra modes are its curvature – but it does say the measurement can distinguish far more shapes than the fit extracts. The limit is how qSLS parameterises the physics, not the information in the record.

Discrimination does not privilege a model. T-optimality [4] asks a different question: which design makes the candidates disagree most? The criterion is

$$T(d) \;=\; \min_{\theta_2} \big\| m_1(\hat{\theta}_1, d) - m_2(\theta_2, d) \big\| / \sigma, \tag{19}$$

and the minimisation is the whole content of it. A rival that can retune its parameters to imitate has not been discriminated against, so the separation must be measured against the rival's *best imitation*, not against its own fitted values.

Two distinct errors were made here, and both inflate T for the same structural reason: T is a *minimum*, so anything that stops short of it – evaluating at fixed parameters, or a minimisation that terminates early – reports discrimination that is not there. An earlier version of this table evaluated each rival at its own fit, giving 2.69 for qSLS–SLS at the present design. Re-fitting the rival with a single Nelder–Mead pass gave 0.67. Restarting and polishing that minimisation changed other entries again: at 1200 μm the reported 4.92 fell to 2.38, and a 0.1% change of design moved one loose evaluation from 7.25 to 1.36, which is the signature of the optimiser rather than the physics. The converged values are:

design	qSLS–SLS	qSLS–qKV	qSLS–NHKV
277 μm, stretch 7.09 (present)	0.67	4.53	4.37
277 μm, stretch 5.00	0.72	4.61	4.61
277 μm, stretch 3.00	0.74	1.72	1.72
199 μm, stretch 3.55 (E-optimal)	0.56	2.59	2.59
895 μm, stretch 4.93	**2.39**	7.13	**7.13**
910 μm, stretch 5.19	1.36	6.97	6.97
1200 μm, stretch 7.09	2.38	5.03	5.63
100 μm, stretch 20.0	1.26	**8.04**	4.17
57 μm, stretch 19.5	1.64	5.53	5.00

At the present design SLS imitates qSLS to within 0.67 noise units, comfortably inside the measurement scatter. That figure is corroborated from an unrelated direction: a systematic offset of 0.668σ over 201 samples contributes $201 \times 0.668^2 = 89.7$ to χ^2, against the measured log-evidence gap of 101.6 between the two models. Two computations sharing no machinery, agreeing to about 12%.

These values are upper bounds, and the criterion is the reason. T is defined by an inner *global* minimisation, and that minimisation is multimodal. Nelder–Mead and Gauss–Newton both converge, to different basins: at 895 μm Nelder–Mead returns 2.39 where Gauss–Newton returns 1.52, and at 1200 μm the ordering reverses. Multistart Gauss–Newton – which analytic Jacobians make 44× cheaper than the derivative-free version, 34 s against 25 minutes for the whole table – finds a lower value in 12 of 27 cells, and a stationarity check built from the same Jacobians reports that 11 cells are still not at a minimum. Every improvement to the optimiser lowers the scores, because a better-optimised rival imitates better. The tabulated values should therefore be read as upper bounds on discrimination, and the true separations are smaller than any of them.

This is not a defect of the implementation but of the criterion. A minimum is attained at a point, so landing in the wrong basin gives a wrong answer with no indication that anything went wrong, and enumerating the outer design space exactly does not repair it: a dense grid over an unreliable integrand produces a front that is precisely wrong. Any criterion built on an argmin – T-optimality, profile likelihood, minimax design – inherits this.

The structural fix is to replace the minimum with an integral. Treating the model label as a discrete parameter makes discrimination an expected-information-gain problem over the augmented parameter (θ, m), whose inner quantity is the marginal likelihood $\int p(Y \mid \theta, M, d)\, \pi(\theta \mid M)\, \mathrm{d}\theta$. An integral accumulates over all modes rather than requiring that the best one be located, so multimodality stops being a failure mode and becomes a property of the integrand; a missed mode biases the evidence slightly instead of corrupting the criterion. It also unifies discrimination with the parameter-precision criteria above in one framework. The cost is that the marginal likelihood is prior-sensitive in a way T is not, so the prior becomes part of the criterion rather than a nuisance.

The criteria do not agree. Computed correctly, the criteria want different experiments. E-optimality peaks at 199 μm and stretch 3.55; separating qSLS from SLS or from NHKV is best at 895 μm and stretch 4.93; separating qSLS from qKV is best at 100 μm and stretch 20, near the opposite corner of the design space. At the E-optimal design the discrimination scores are the worst in the table, so the conflict is not marginal.

The reason is physical rather than numerical. Distinguishing a material with a relaxation mode from one without requires a collapse fast enough to excite that mode, whereas separating g from α requires a gentle collapse, since the term of Z_e carrying that distinction dies as λ^{-4}. The design cannot be both fast and gentle.

There is accordingly no single best design, and a scalar design objective is answering a question that has no scalar answer. What replaces it is the *Pareto front*: the set of designs on which no other design is better on every criterion at once. Enumerating a 14×18 grid over $(R_{\max}, \lambda)$ and applying that test leaves 12 designs undominated, spanning 50–1200 μm and stretch 4–18. Reporting that set, rather than a single optimum, is the honest summary of an experiment that must serve several questions at once; `pyimr.pareto.explore_tradeoff` computes it. Given the caution above, the T columns entering the domination test are upper bounds, so the front should be regarded as provisional in its discrimination coordinates while the E-optimality coordinate – an eigenvalue of $J^{\mathsf{T}}J$, with no inner optimisation – is not subject to that reservation.

Discrimination as an integral rather than a minimum

Everything above scores a design by T, which is defined through a *global minimum* over the rival's parameters. That definition is the source of the trouble in the previous subsection, and the trouble is structural rather than computational. This subsection derives a criterion with the minimum replaced by an integral, states precisely why that is more robust, and reports what it gives.

The model label is a parameter. Write $M \in \{M_1, M_2\}$ for the model index with prior $p(M)$, and let each model carry its own parameter θ with prior $\pi(\theta \mid M)$. Nothing distinguishes M from any other unknown, so the information a design d yields about it is the mutual information

$$I(M; Y \mid d) = \mathbb{E}_M \, \mathbb{E}_{Y|M,d} \left[\log \frac{p(Y \mid M, d)}{p(Y \mid d)} \right], \qquad p(Y \mid d) = \sum_M p(M) \, p(Y \mid M, d), \tag{20}$$

which is the expected reduction in entropy of M, exactly as the expected information gain used for the design study above is the expected reduction in entropy of θ. Discrimination and estimation are therefore the same criterion applied to different components of the augmented unknown (θ, M); they are not competing philosophies, and the earlier tension between T- and E-optimality is a tension between *which component we care about*, not between two schools.

The quantity entering (20) is the marginal likelihood, or evidence,

$$p(Y \mid M, d) = \int p(Y \mid \theta, M, d) \, \pi(\theta \mid M) \, \mathrm{d}\theta, \tag{21}$$

the same object as (4) but now at a hypothetical design and for hypothetical data. It is an integral over θ, and that single fact is what the rest of this subsection turns on.

The criterion actually computed. Rather than (20) we score a design by the expected log Bayes factor in favour of the model that generated the data,

$$U(d) = \mathbb{E}_{Y \sim p(\cdot|M_1,d)} \left[\log p(Y \mid M_1, d) - \log p(Y \mid M_2, d) \right] = D_{\mathrm{KL}}(p(\cdot \mid M_1, d) \, \| \, p(\cdot \mid M_2, d)) \, . \tag{22}$$

Recognising it as a Kullback–Leibler divergence between the two prior-predictive distributions settles its properties without further work. By Gibbs' inequality $U(d) \geq 0$, with equality if and only if the two predictives agree almost everywhere: *a model cannot discriminate against itself, at any design.* That is a sharp, falsifiable statement about the estimator, and it is the property the implementation is tested against – two independently drawn banks from one family must score zero, and any systematic excess is bias. Related to (20) by $I(M; Y \mid d) = \frac{1}{2}[D_{\mathrm{KL}}(p_1 \| \bar{p}) + D_{\mathrm{KL}}(p_2 \| \bar{p})]$ with $\bar{p}$ the equal mixture, so (22) is the same comparison read against the rival rather than against the mixture, and is the more direct statement of "how strongly will this experiment favour the truth".

Why an integral is robust where a minimum is not. Both criteria require an inner computation that can go wrong. The asymmetry is in how the error propagates.

For T, any optimiser returns $\hat{T} = \|m_1 - m_2(\hat{\theta}_2)\|/\sigma$ at whatever point it stopped, and since T is a minimum, $\hat{T} \geq T$ always. If $\hat{\theta}_2$ lies in the correct basin but is imprecise, stationarity gives $\hat{T} - T = O(\|\hat{\theta}_2 - \theta_2^\star\|^2)$ and the error is negligible. If it lies in the *wrong basin*, the error is $O(1)$: a fixed, unbounded overstatement with nothing in the returned value to indicate it. This is what was measured – Nelder–Mead and Gauss–Newton returning 2.39 and 1.52 at one design and reversing at another.

For the evidence, suppose the posterior has modes $k = 1, \ldots, K$ contributing $Z = \sum_k Z_k$, and a procedure finds only a subset $\mathcal{S}$. Then

$$\log \hat{Z} - \log Z = \log\left(1 - \sum_{k \notin \mathcal{S}} \frac{Z_k}{Z}\right), \tag{23}$$

so the error is controlled by the *posterior weight* of what was missed. Missing a subdominant mode costs $O(Z_k/Z)$ and missing a negligible one costs nothing. Because the contributions add, a multistart that finds several modes can sum them rather than choose between them, and each additional mode found monotonically improves the estimate. There is no analogue for a minimum, where finding more candidates only ever changes which single one is reported.

This is the whole argument for the change, and it is worth stating plainly: multimodality is a *failure mode* for a minimum and merely a *property of the integrand* for an integral.

Estimating the evidence I: prior sampling, and why it fails here. The direct estimator draws $\theta_s \sim \pi(\theta \mid M)$ and averages the likelihood,

$$\hat{Z} = \frac{1}{S}\sum_{s=1}^{S} p(Y \mid \theta_s, M, d), \qquad w_s = p(Y \mid \theta_s, M, d), \tag{24}$$

which is unbiased for Z, though $\log \hat{Z}$ is biased low by Jensen's inequality. Its reliability is carried by the spread of the weights, summarised by the effective sample size

$$S_{\text{eff}} = \frac{\left(\sum_s w_s\right)^2}{\sum_s w_s^2} \;\in\; [1, S], \tag{25}$$

which equals S when all draws contribute equally and 1 when a single draw dominates. It costs nothing to compute and it is the only thing standing between this estimator and a confidently reported number with no content.

That it must fail here can be seen in advance. With N samples the log-likelihood scales as $\log p(Y \mid \theta) \simeq -\frac{1}{2}N\bar{d}(\theta)$ for a mean squared discrepancy $\bar{d}$, so weights are $w_s \propto \exp[-\frac{1}{2}N\bar{d}(\theta_s)]$ and the ratio of the two largest is exponential in N. Useful draws are those falling inside the posterior, whose width is $O(N^{-1/2})$ in each of p directions, so the expected number of them is

$$\mathbb{E}[\#\text{useful}] \;\sim\; S \cdot \frac{\text{vol(posterior)}}{\text{vol(prior)}} \;\sim\; S\,N^{-p/2}. \tag{26}$$

For the present study $N = 201$, $p = 4$ and $S = 220$, giving $220 \times 201^{-2} \approx 5 \times 10^{-3}$: fewer than one useful draw. The measurement agrees exactly – $S_{\text{eff}} = 1.0$ at *every* design tried, with reported values between 793 and 10 979 nats that are pure noise. The standard errors were 56 to 130 throughout and looked entirely respectable. Without (25) the failure is invisible.

Estimating the evidence II: Laplace. Expanding $\log[p(Y \mid \theta)\pi(\theta)]$ to second order about its maximum $\hat{\theta}$ and performing the resulting Gaussian integral gives, as in (5),

$$\log Z \simeq \log p(Y \mid \hat{\theta}) + \log \pi(\hat{\theta}) + \tfrac{p}{2}\log 2\pi - \tfrac{1}{2}\log\det H, \qquad H = -\nabla^2 \log p(Y \mid \theta)\big|_{\hat{\theta}}. \tag{27}$$

For Gaussian data $H = J^{\mathsf{T}}J + \sum_i r_i \nabla^2 r_i$, and dropping the second term – the Gauss–Newton approximation, valid when residuals are small or the model nearly linear – leaves $H \simeq J^{\mathsf{T}}J$, already computed by the fit. Parameterising so the prior is uniform on the unit cube makes $\pi(\hat{\theta}) = 1$, and with whitened residuals $r = (m - y)/\sigma$,

$$\log Z \simeq -\tfrac{1}{2}\|r\|^2 - \tfrac{N}{2}\log\left(2\pi\sigma^2\right) + \tfrac{p}{2}\log 2\pi - \tfrac{1}{2}\log\det\left(J^{\mathsf{T}}J\right). \tag{28}$$

The first two terms are the best fit; the last two are the Occam factor, and their sign is what prevents a more flexible model from winning on fit alone. Crucially for the present purpose, (28) does not care how sharp the likelihood is, which is precisely the regime that destroyed (24). It is affordable only because the fit uses analytic Jacobians: derivative-free minimisation of the same objective costs 55 s against 7.6 s, and the Jacobian needed for $\det(J^{\mathsf{T}}J)$ is a by-product of the solve rather than an extra cost.

The two criteria are consistent where both can be computed. The criteria are not independent, and the relation between them is a check on both. If the true model fits to within noise, $\chi_1^2 \approx N$, while a rival that cannot fit leaves a residual of root-mean-square $T\sigma$, so $\chi_2^2 \approx NT^2$. Substituting into the leading term of (28),

$$\log \text{BF} \;\approx\; \tfrac{1}{2}\left(\chi_2^2 - \chi_1^2\right) \;\approx\; \tfrac{N}{2}\left(T^2 - 1\right) \;\xrightarrow{T \gg 1}\; \tfrac{N}{2}T^2, \tag{29}$$

up to the Occam terms, which T has no way to express. Testing (29) against the measured values:

design	T (vs SLS)	$\frac{N}{2}T^2$ predicted	measured U (nats)
199 μm, stretch 3.55	0.56	31.5	35.4 ± 12.2
895 μm, stretch 4.93	1.52	232	150 ± 73

Two computations sharing no machinery – one a constrained least-squares minimum, the other a marginal likelihood by Laplace – agreeing to within their stated uncertainty. The residual gap at 895 μm is of the right sign and size for the Occam penalty that qSLS pays for its fourth parameter.

What it costs, and where it is not yet trustworthy. Two prices are paid for removing the minimum, and both are real.

Prior sensitivity. Z depends on $\pi(\theta \mid M)$ through the normalisation of (21), and widening a prior by a factor c in each of p directions divides the evidence by roughly c^p without changing the fit at all. This is the Lindley–Jeffreys [5, 6] effect, and it is not a defect to be engineered away: a model whose prior is diffuse really is less supported by data that pin it down. But it means the prior is part of the criterion and must be stated, where T needed no prior. Centring each rival's prior on its fit to the *existing* record produced a criterion that measured prior placement instead of discriminability, with every Laplace expansion invalid; the results reported here use the wide, physically motivated box of the model-selection grid, identical across models on every shared parameter.

Boundary pinning. Equation (27) presumes an interior maximum. In 50–90% of the fits at each design, the rival's best parameters lie on a face of the prior box, where the expansion does not apply and the reported value is not to be trusted. That this happens is itself informative – a rival that can imitate qSLS only by driving a parameter to the edge of physical plausibility has, in a meaningful sense, been discriminated against – but expressing that requires a criterion built for a constrained maximum rather than a quadratic expansion about an interior one. Determining which parameter pins, and whether the pinning is physical or an artefact of the box, is the outstanding step.

The honest summary is that the integral criterion is better founded than T and not yet more reliable on this problem. The multimodality that made T untrustworthy is genuinely gone; what replaces it is a dependence on the prior, now explicit rather than hidden, and a boundary condition that the current estimator does not handle.

Which criterion the conflict is between

The scaling argument of Section 2.7 says that a design change which merely amplifies the signal leaves the condition number κ untouched, so degeneracy is a property of the eigenvectors and only a design that rotates them can remove it. The consequence for these records is that E-optimality preferred a larger bubble (895 μm, $\lambda_{\min} = 12.8$) while conditioning and eigenvector composition preferred a gentler collapse at the present size (277 μm at stretch 5, $\kappa = 4.2 \times 10^3$, with the modulus–stiffening share of the worst direction falling from 87% to 66%). Both are correct answers to different questions, and (15) says which question matters here.

So there are two distinct disagreements in play, and they have different resolutions. The classical criteria disagree among themselves because they summarise one spectrum differently; that is a choice about what one wants to learn. Estimation and discrimination disagree because they are about different quantities altogether. The next subsection resolves the second, and in doing so dissolves the first.

Designing over measures, and certifying the answer

Every design result above shares a defect that no amount of care in computing them repairs: each is the output of a search over a single design point, that search is not convex, and nothing in its output distinguishes a good answer from a bad one. The evidence that this is not hypothetical is in the preceding subsections. A surrogate search recovered 16% of the best conditioning actually available and reported the result without qualification; a dense grid over a criterion whose inner minimisation was unreliable produced a front that was precise and wrong. Both look exactly like success.

The classical formulation removes the defect rather than mitigating it.

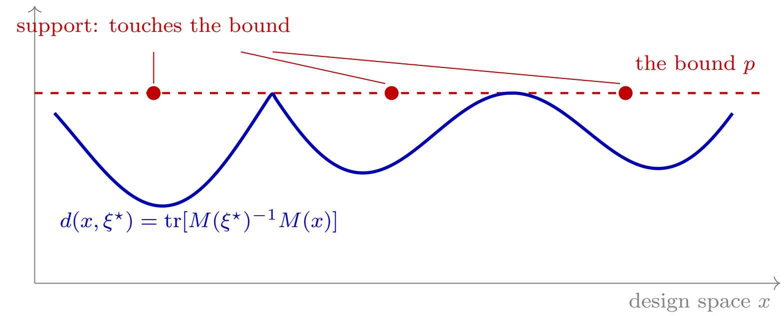

Figure 13: The certificate, which is what makes an optimal design a proof rather than a stopping rule. Kiefer and Wolfowitz: a measure $\xi^\star$ is D-optimal *if and only if* its sensitivity $d(x, \xi^\star)$ never exceeds the parameter count p anywhere in the design space, and attains p exactly on the points carrying weight. Both halves are visible at once – a curve capped by a level it touches and never crosses – so a search reports not merely that it stopped improving but that there is nowhere left for a better design to hide. The gap $\max_x d(x, \xi^\star) - p$ is the proof's residue; 10^{-9} in the designs here.

Designs as measures. Instead of choosing one design x from a candidate set $\mathcal{X}$, choose a probability measure ξ on $\mathcal{X}$: run a fraction ξ_i of the bubbles at setting x_i. This is not an abstraction imposed for mathematical convenience – it is what an experimenter with a run budget does anyway, and it is strictly more expressive, since a single design is the special case of a measure concentrated on one point. The information available from a run allocated by ξ is the average

$$M(\xi) \;=\; \sum_i \xi_i \, M(x_i), \qquad M(x) = J(x)^\mathsf{T} J(x), \tag{30}$$

because information from independent runs adds and ξ says how many of each to take.

The consequence is the whole point: $M(\xi)$ is *linear* in ξ.

Concavity, and why the first-order condition is enough. $\log\det$ is concave on the positive-definite cone, and the composition of a concave function with a linear map is concave, so $\Phi(\xi) = \log\det M(\xi)$ is concave on the simplex – a convex set. A concave function on a convex set has no local maximum that is not global, and its first-order condition is not merely necessary but sufficient. Every pathology this study ran into – wrong basins, surrogate blind spots, answers that improve whenever the optimiser does – is a symptom of non-convexity, and none of them can occur here.

The same holds for $\Phi_A = -\operatorname{tr} M(\xi)^{-1}$ and $\Phi_E = \lambda_{\min}(M(\xi))$, and for any non-negative combination of concave criteria.

The directional derivative. Move mass α from ξ toward the single design x, giving $\xi_\alpha = (1-\alpha)\xi + \alpha\,\delta_x$. By linearity $M(\xi_\alpha) = (1-\alpha)M(\xi) + \alpha M(x)$, and using $\partial \log\det A = \operatorname{tr}(A^{-1}\partial A)$,

$$\phi(x, \xi) \;\equiv\; \frac{\partial}{\partial\alpha} \log\det M(\xi_\alpha)\bigg|_{\alpha=0} = \operatorname{tr}\big[M(\xi)^{-1} M(x)\big] - p. \tag{31}$$

For a linear utility u averaging over the design, the same construction gives $\phi_u(x, \xi) = u_x - u^\mathsf{T}\xi$.

The General Equivalence Theorem. Since Φ is concave and the simplex is the convex hull of its vertices δ_x, the measure $\xi^\star$ maximises Φ if and only if no vertex direction ascends:

$$\boxed{\xi^\star \text{ is D-optimal} \iff d(x, \xi^\star) \equiv \operatorname{tr}\big[M(\xi^\star)^{-1} M(x)\big] \le p \quad \text{for all } x \in \mathcal{X},} \tag{32}$$

with equality at every support point of $\xi^\star$. This is the theorem of Kiefer and Wolfowitz [7], and it is the object this whole section exists to obtain: a quantity that can be computed from a candidate answer and that *proves* the answer optimal. The gap $\max_x d(x, \xi) - p$ is a certificate, not a convergence report.

Two further facts come free. The equality condition says the support sits exactly where $d(x, \xi^\star)$ attains its maximum, so the optimal design places its runs at the settings of greatest standardised prediction variance – D-optimality and G-optimality coincide. And by Carathéodory's theorem applied to the convex hull of the $M(x)$, an optimal measure supported on at most $p(p+1)/2+1$ points always exists: at most 11 here.

Computing it. The implementation uses the multiplicative algorithm of Silvey, Titterington and Torsney [8],

$$\xi_i \longleftarrow \frac{\xi_i\, F_i}{\sum_j \xi_j F_j}, \qquad F_i = \mathrm{tr}\big[M(\xi)^{-1} M(x_i)\big] \;\; \text{for} \;\; \Phi = \log\det, \tag{33}$$

which stays on the simplex by construction – weights remain non-negative and normalised without projection – and increases the criterion at each step. Note that $\sum_i \xi_i F_i = p$ identically, so (33) moves mass toward designs whose $d(x, \xi)$ exceeds p, which is precisely the set that (32) says must be empty at the optimum. The update and the certificate are the same quantity.

Vertex-direction steps with a line search were tried first and cannot converge here: the step required near the optimum falls below any fixed search grid and the iteration stalls with a positive gap. Convergence of (33) degrades as candidates crowd together, since a design just off the support is nearly as good as one on it and its weight decays in proportion – certifying a parabola on $[-1, 1]$ took 6400 iterations over 51 candidates, 25 000 over 101, and 329 000 over 401, with the criterion value correct to six figures long before any of those. An exhausted iteration budget therefore yields a converged value without a proof, which is a different thing from a wrong answer, and the reported gap distinguishes them.

The certified design for these records. Over 480 candidates spanning $R_{\max} \in [50, 1200]\,\mu$m and stretch $[3, 20]$, with $M(x) = J^\mathsf{T} J$ for the four qSLS parameters at the posterior median, the algorithm certifies at a gap of 9×10^{-10} in 1406 iterations: $\max_x \mathrm{tr}[M^{-1} M(x)] = 4.00000000$ against $p = 4$. The optimal measure is

$R_{\max}$	stretch	weight
1015 μm	6.70	0.470
161 μm	19.26	0.303
50 μm	17.78	0.227

against a best single design of 1015 μm at stretch 6.70. The measure exceeds it by 6.99 in $\log\det$, a D-efficiency ratio of $\exp(6.99/4) = 5.74$; equivalently the confidence ellipsoid is $\exp(6.99/2) \approx 33$ times smaller in volume, and a typical parameter's interval is $\exp(6.99/8) = 2.4$ times tighter. No single design can match it, and this is not a claim about the search: (32) certifies that none exists.

The mixture is the resolution of the conflict. The support is worth reading physically. Nearly half the runs go to a large, gently driven bubble and the rest to small, hard-driven ones – the two corners that the earlier subsections found pulling in opposite directions. Separating g from α wants a gentle collapse because the distinguishing term of Z_e decays as λ^{-4}; exciting the relaxation mode wants a fast one. A single design has to choose between them and a measure does not, so the trade-off that motivated the Pareto front dissolves once the design space is enlarged to the one the experiment actually admits. The front was answering a question made artificially hard by the restriction to a single setting.

Discrimination in the same frame. A utility that averages over runs – an expected log Bayes factor per run, as in (22), since log evidence is additive over independent experiments – is linear in ξ and therefore concave, so the compound criterion

$$\Phi_\beta(\xi) = (1-\beta) \log\det M(\xi) + \beta\, u^\mathsf{T} \xi, \qquad \beta \in [0, 1], \tag{34}$$

is concave for every β and inherits the certificate unchanged; its directional derivative is the corresponding combination of (31) and $u_x - u^\mathsf{T}\xi$. This is the compound-criterion construction of Atkinson and of Cook and Wong [9, 10], and it is the principled form of the estimation-versus-discrimination trade this study has otherwise handled by search. Sweeping β traces the Pareto front, and because each criterion is concave in ξ the front is convex – which is why the classical literature scalarises with weighted sums where this study needed a Chebyshev form. The non-convexity that forced the more elaborate scalarisation was an artefact of optimising over a single design point.

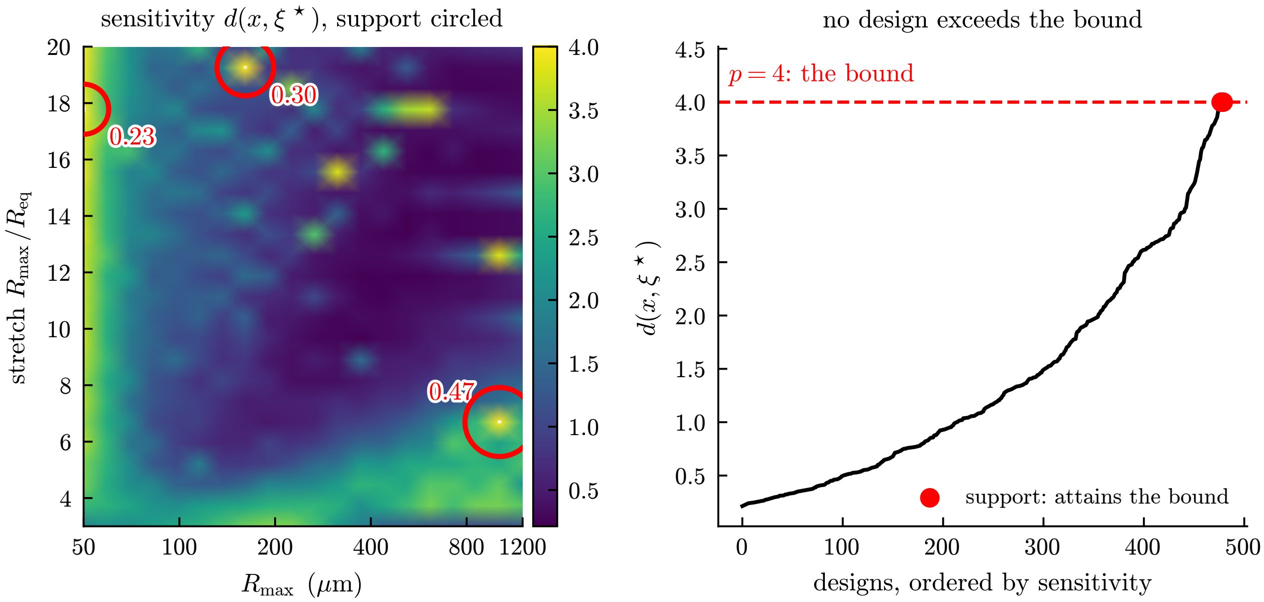

Figure 14: The equivalence-theorem certificate. Left: the sensitivity $d(x, \xi^\star) = \mathrm{tr}[M(\xi^\star)^{-1}M(x)]$ over the design plane, with the three support points circled and labelled by weight; the dashed contour is the level $p = 4$. Right: the same values sorted. No design exceeds the bound and the support attains it exactly, which is the proof – $\max_x d = 4.00000000$ against $p = 4$, gap 9×10^{-10}. Nothing analogous exists for the single-design searches elsewhere in this document.

What this does not fix. Three limitations are worth stating plainly. The information matrices are evaluated at a point estimate of θ, so the design is locally optimal in the parameters even though it is globally optimal in ξ; a prior average over θ restores the guarantee at proportionate cost, since the average of concave criteria is concave. The candidate set is still a grid, so the certificate is a proof of optimality *over that grid* – though the sensitivity function $d(x, \xi^\star)$ can be evaluated anywhere, which turns checking for a missed region into a cheap scan rather than another search. And the compound form was not run on these records, because the per-design log Bayes factor still suffers the boundary pinning described above.

8 The temperature trend

The most informative result is not any single number but a trend (Figure 16). The advantage that the stiffening term buys over pure relaxation shrinks monotonically as the gel warms: the gap between `SLS` and `qSLS` falls from 2.94 versus 1.35 at 15 °C to 1.41 versus 1.23 at 33 °C. A warmer, softer gel needs less strain-stiffening to explain its record. Because this is a trend across three independent datasets rather than a single fitted value, it is harder to attribute to a fitting artefact than any individual χ^2.

9 Limitations

The dropped points are outside the model, not outside the solver. Roughly one point in six of the winner's parameter box cannot be integrated. That is not a numerical shortcoming. In that region the model expands to $R/R_0 = 2132$ after its collapse – identically at relative tolerances of 10^{-3}, 10^{-4} and 10^{-5}, so a converged property of the equations rather than an artefact. The bubble begins at its maximum radius and its energy is finite, so growth of that size is the constitutive model failing, and refusing those points is the correct outcome.

The count of physically admissible points is pinned near 1958 of 2401 sampled whatever the settings: looser tolerances and a 20× step budget convert failures into runaways, never into usable solves. This corrects an earlier reading of the same failures as a stiffness problem. It also means the dropped points cannot be recovered by numerical effort, and that no claim about the winner's fit should be made in that region.

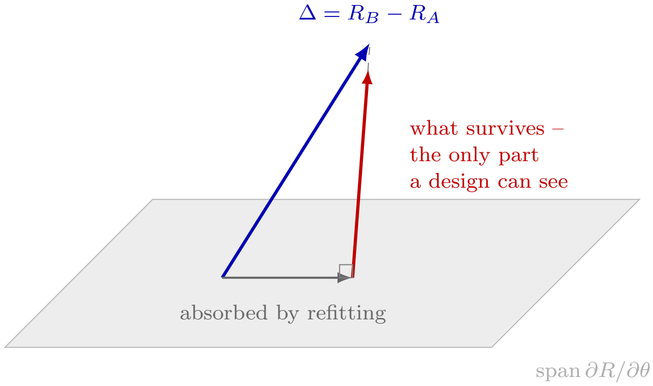

Figure 15: Why a model difference can be invisible at every design. Changing a model perturbs the trace by $\Delta = R_B - R_A$. The component lying in the span of the material sensitivities is reproduced by adjusting $g, \mu, \lambda_1, \alpha$ – refitting absorbs it, and no experiment recovers it. Only the orthogonal remainder is evidence that the model changed. Measured on the 15 °C record and reported in Section 6: the bubble-dynamics operator is 77.7% absorbed, leaving under two noise units, which is why its ranking proved so fragile; the constitutive and thermal axes leave 8.1 and 8.5.

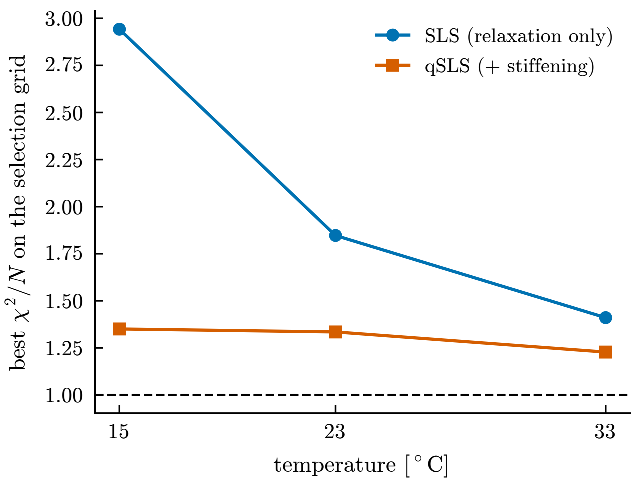

Figure 16: What the stiffening term buys, against temperature. The advantage of `qSLS` over `SLS` narrows as the gel warms.

The ranking survives; the parameter estimates do not. Independent sequential Monte Carlo runs give the winner 2173.55 ± 0.19 in log evidence against 2071.90 ± 0.38 for the relaxation-only model – a gap of 101.6 against run-to-run noise near 0.4, so the comparison is resolved by a wide margin. The tensor grid cannot say this on its own: it returns a single number with no uncertainty.

The same grid cannot estimate parameters. Refining it from 12 to 20 points per axis moves the log evidence by 17.3 and moves the best-fit point from $g = 658$, $\alpha = 1.52$ to $g = 144$, $\alpha = 8.86$, because the best cell is simply whichever node lands nearest the ridge. Refinement cannot fix this: the posterior standard deviations are 0.0018, 0.0025, 0.0055 and 0.078 in unit coordinates against a grid spacing of 0.053, so the posterior occupies about 10^{-7} of the prior volume and a uniform grid would need of order 10^{10} points. Discretisation error of 17.3 sits well inside the 101.6 gap, so the ranking is safe; the reported best-fit parameters are not. Figure 18 shows which models lose points.

Consequences for the numbers reported here. Three methods that resolve the posterior – sampling, a refined grid, and continuous optimisation – agree that g lies near 10^2 Pa. The coarse grid used for the ranking reports 2154 Pa. The ranking does not depend on this, but any material property quoted from the grid does. We also note that a modulus of that size is implausibly soft for gelatin, which is the degeneracy above making

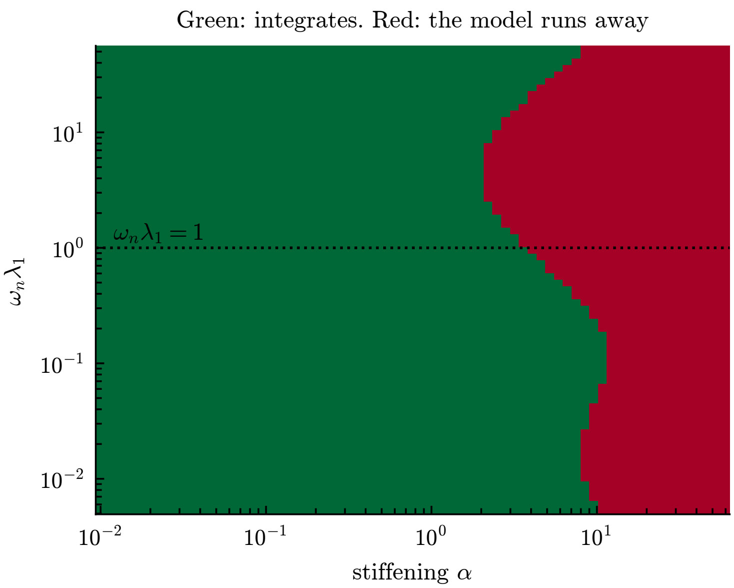

Figure 17: Where the winner can be integrated, at the fitted modulus and viscosity. Failure is governed principally by the stiffening, and the threshold falls to its lowest where the relaxation time is comparable to the bubble's own period – but it is a boundary that moves, not an isolated band.

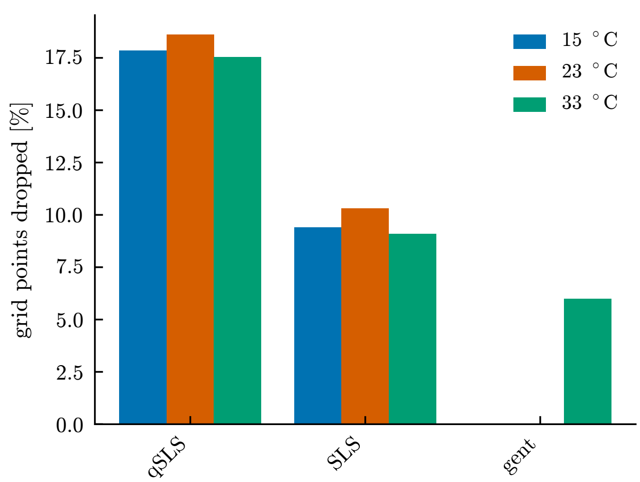

Figure 18: Fraction of each model's parameter grid that could not be integrated. Models not shown lost no points.

itself felt: the fit buys a lower residual by trading modulus against stiffening, along a direction the data barely constrain.

One candidate was never genuinely exercised. The Gent solid appears in the comparison but cannot be tested on these records. Its energy diverges as $I_1 - 3 \to J_m$, and a Gent solid soft enough to be distinguishable from Neo-Hookean predicts a collapse it cannot sustain, so it locks up. Where it does integrate, it differs from Neo-Hookean by 1.7×10^{-4} of $R_{\max}$ against measurement noise near 2×10^{-2} – two orders of magnitude below what the data can resolve. Any statement about strain-stiffening laws here excludes it.

A Statistical background

This appendix develops, from first principles, the results the main text uses. It is self-contained and assumes only calculus and elementary probability.

A.1 Inference as conditioning

Write θ for parameters, y for data, M for a model. The product rule $p(\theta, y) = p(y \mid \theta)\, p(\theta) = p(\theta \mid y)\, p(y)$ rearranges to Bayes' theorem,

$$p(\theta \mid y, M) = \frac{p(y \mid \theta, M)\, \pi(\theta \mid M)}{p(y \mid M)}, \qquad p(y \mid M) = \int p(y \mid \theta, M)\, \pi(\theta \mid M)\, \mathrm{d}\theta. \tag{35}$$

The denominator is a normalising constant in θ, which is why it is ignored when estimating parameters and why it is the entire content of model comparison: applying (35) at the level of models gives

$$\frac{p(M_1 \mid y)}{p(M_2 \mid y)} = \underbrace{\frac{p(y \mid M_1)}{p(y \mid M_2)}}_{\text{Bayes factor } B_{12}} \cdot \frac{p(M_1)}{p(M_2)}. \tag{36}$$

Everything the data say about which model to prefer is in B_{12}. Jeffreys' conventional reading takes $\log B > 1$ as substantial and $\log B > 5$ as decisive; the gap reported here is over 100, which is why its numerical resolution had to be checked rather than assumed.

A.2 The score, and two definitions of information

Let $\ell(\theta) = \log p(y \mid \theta)$. The *score* is $u = \nabla_\theta \ell$. Since $\int p(y \mid \theta)\, \mathrm{d}y = 1$ for every θ, differentiating under the integral gives

$$\mathbb{E}_\theta[u] = \int \frac{\nabla p}{p}\, p\, \mathrm{d}y = \nabla \int p\, \mathrm{d}y = 0, \tag{37}$$

so the score has zero mean at the true parameter. Its covariance defines the Fisher information, $F = \mathbb{E}[uu^\mathsf{T}]$. Differentiating the same identity twice gives the equivalent and more useful form

$$F = \mathbb{E}\left[-\nabla^2 \ell\right], \tag{38}$$

the expected curvature. The two agree because $\nabla^2 \log p = \nabla^2 p/p - (\nabla p/p)(\nabla p/p)^\mathsf{T}$ and the first term integrates to zero. Curvature and variance-of-score are the same object, which is the reason a sharply peaked likelihood and a precisely determined parameter are the same statement.

A.3 The Cramér–Rao bound

For an unbiased estimator $\hat{\theta}(y)$, differentiate $\mathbb{E}[\hat{\theta}] = \theta$ under the integral to get $\mathbb{E}[\hat{\theta}\, u^\mathsf{T}] = I$. Applying the Cauchy–Schwarz inequality to the pair $(\hat{\theta} - \theta,\ u)$ in the scalar case gives $\mathrm{Var}(\hat{\theta})\, \mathbb{E}[u^2] \geq 1$, i.e. $\mathrm{Var}(\hat{\theta}) \geq 1/F$; the multivariate statement is

$$\mathrm{Cov}(\hat{\theta}) \succeq F^{-1}. \tag{39}$$

So F^{-1} is a floor on achievable precision, attained asymptotically by maximum likelihood: $\sqrt{N}(\hat{\theta}_N - \theta) \to \mathcal{N}(0, F_1^{-1})$ under regularity. This is the licence for reading F^{-1} as a covariance and its eigenvectors as the axes of a confidence ellipsoid – and the regularity conditions are exactly what near-singular F threatens, which is why sloppiness is worth detecting rather than tolerating.

A.4 Gaussian data, and why $F = J^\mathsf{T} J$

With independent Gaussian observations of known deviation, $\ell = -\frac{1}{2}\sum_i (y_i - m_i(\theta))^2/\sigma_i^2 + \text{const}$. Then

$$\frac{\partial \ell}{\partial \theta_j} = \sum_i \frac{y_i - m_i}{\sigma_i^2}\frac{\partial m_i}{\partial \theta_j}, \qquad -\frac{\partial^2 \ell}{\partial \theta_j \partial \theta_k} = \sum_i \frac{1}{\sigma_i^2}\left[\frac{\partial m_i}{\partial \theta_j}\frac{\partial m_i}{\partial \theta_k} - (y_i - m_i)\frac{\partial^2 m_i}{\partial \theta_j \partial \theta_k}\right]. \tag{40}$$

Taking the expectation kills the second bracket because $\mathbb{E}[y_i - m_i] = 0$, leaving $F = J^\mathsf{T} J$ with $J_{ij} = \sigma_i^{-1}\partial m_i/\partial \theta_j$. Note what was discarded: the curvature of the model. At the optimum the residuals are small and the Gauss–Newton form is accurate; far from it, or with a badly misspecified model, F and the observed Hessian differ, and the discrepancy is itself a diagnostic.

A.5 The Laplace approximation and the Occam factor

To evaluate $Z = \int e^{\ell(\theta)}\pi(\theta)\,\mathrm{d}\theta$, expand about the maximum $\hat{\theta}$ where $\nabla \ell = 0$:

$$\ell(\hat{\theta} + \delta) = \ell(\hat{\theta}) - \tfrac{1}{2}\delta^\mathsf{T} H \delta + O(\|\delta\|^3), \qquad H = -\nabla^2 \ell(\hat{\theta}). \tag{41}$$

Holding π constant over the peak and using $\int e^{-\delta^\mathsf{T} H \delta/2}\mathrm{d}\delta = (2\pi)^{p/2}|H|^{-1/2}$,

$$Z \simeq e^{\ell(\hat{\theta})}\,\pi(\hat{\theta})\,(2\pi)^{p/2}\,|H|^{-1/2}. \tag{42}$$

The correction is $O(N^{-1})$ relative for regular problems. Writing the posterior volume as $V_{\text{post}} = (2\pi)^{p/2}|H|^{-1/2}$ and the prior density as $1/V_{\text{prior}}$ where it is flat, the second factor is $V_{\text{post}}/V_{\text{prior}}$: the fraction of the space the data leave standing. A model with more parameters has a larger V_{prior} and pays for it unless the fit improves enough to compensate. Nothing was added by hand; the penalty is the Jacobian of the integral.

Taking logarithms of (42), using $H \approx N F_1$ for N independent observations, and dropping terms that do not grow with N,

$$-2\log Z \simeq -2\ell(\hat{\theta}) + p \log N + O(1), \tag{43}$$

which is the Bayesian information criterion. BIC is therefore not an alternative to the evidence but a coarse asymptotic of it, valid when the posterior is Gaussian, the parameters identified, and N large. AIC answers a different question – expected out-of-sample Kullback–Leibler divergence – and is not an approximation to Z at all. Where a model is sloppy, H is near-singular, $|H|^{-1/2}$ is enormous, and the asymptotic form fails while the integral remains well defined.

A.6 Capping the Occam factor at the prior

The failure just noted is not merely a loss of accuracy; it has a sign, and the sign is wrong. Work in coordinates where the prior is uniform on the unit cube, so $V_{\text{prior}} = 1$ and $H = J^\mathsf{T} J$. The Occam factor is a product over the eigendirections of $J^\mathsf{T} J$,

$$\frac{V_{\text{post}}}{V_{\text{prior}}} = \prod_{i=1}^{p}\sqrt{\frac{2\pi}{\lambda_i}}, \tag{44}$$

each factor being the posterior width in that direction against a prior of width one. A direction the data does not determine has $\lambda_i \to 0$ and contributes a factor diverging to $+\infty$. The evidence then *increases* with the addition of a parameter that does nothing, which is the opposite of what the Occam factor exists to do.

The pathology is entirely an artefact of the Gaussian approximation. The fitted Gaussian has width $\sqrt{2\pi/\lambda_i} \gg 1$ in that direction and so extends far outside the cube the prior actually occupies, while the true

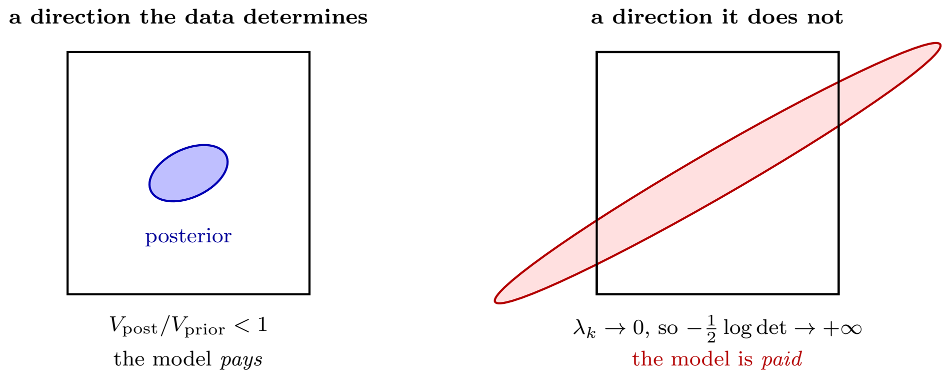

Figure 19: The Occam factor is the share of the prior the data leaves standing, $\prod_k \sqrt{2\pi/\lambda_k}$ over the eigenvalues of $J^\mathsf{T} J$. Left: every direction is narrower than its prior, so a flexible model pays for volume it does not use. Right: a direction the data does not determine has $\lambda_k \to 0$, the fitted Gaussian extends far outside the box the prior occupies, and the expansion returns a factor *greater* than one – the evidence rises when a useless parameter is added. Capping each direction at unity, (45), restores what the geometry already required; on the models of Section 4.4 the uncapped form favoured the two-mode law by 29.7 nats exactly where its second arm did nothing.

integral, taken over the cube, simply returns the prior mass. Imposing what the geometry already requires – a posterior cannot be wider than the prior it came from – gives

$$\log \frac{V_{\mathrm{post}}}{V_{\mathrm{prior}}} = \sum_{i=1}^{p} \min\left(0,\ \tfrac{1}{2}\log\frac{2\pi}{\lambda_i}\right), \tag{45}$$

so a direction the data cannot see costs nothing and buys nothing. Where every $\lambda_i > 2\pi$ each minimum selects its second argument and (45) reduces identically to the logarithm of (44): the cap is a strict generalisation, altering an answer only where the uncapped form was not entitled to one. Section 4.4 gives the size of the error it removes, which on the models considered there is 29.7 nats.

This matters wherever models of differing dimension are compared and some parameters are weakly identified – which, for the sloppy spectra of Section A.7, is the normal case rather than the exceptional one.

A.7 Sloppy models

Write the eigen-decomposition $F = V\Lambda V^\mathsf{T}$ with $\lambda_1 \geq \cdots \geq \lambda_p$. The misfit near the optimum is $\Delta\chi^2 = \sum_k \lambda_k c_k^2$ for $\delta = \sum_k c_k v_k$, so moving a distance c along v_k costs $\lambda_k c^2$ and the 95% contour lies at $c_k = \sqrt{3.84/\lambda_k}$. Multiparameter models of this kind generically show spectra spanning many decades, with a few stiff directions and many sloppy ones – the observation of Gutenkunst [11], Sethna and co-workers, and the reason parameter values in such models are often far less determined than their predictions.

The distinction to keep is between a rank-deficient F, where a direction is *structurally* unidentifiable and no experiment helps, and a merely ill-conditioned F, where the direction is determined but weakly. In logarithmic parameters, an eigenvector is a power-law combination, and the invariant it leaves fixed is read off its components.

A.8 Correlated errors

Let r be the residual vector with covariance C. The Gaussian log-likelihood is

$$-2\ell = r^\mathsf{T} C^{-1} r + \log|C| + N\log 2\pi, \tag{46}$$

and with the Cholesky factorisation $C = LL^\mathsf{T}$ this becomes $\|L^{-1}r\|^2 + 2\sum_k \log L_{kk} + \text{const}$: whitening turns generalised least squares back into ordinary least squares on transformed residuals. Ignoring the off-diagonal

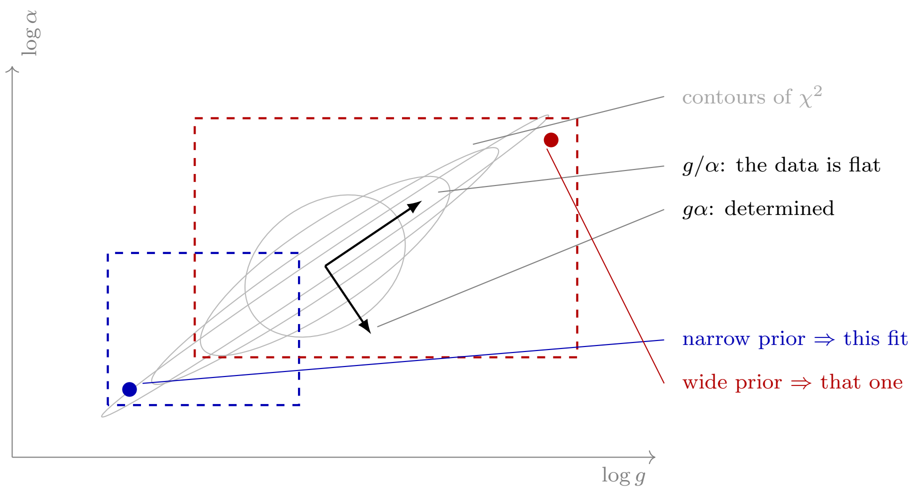

Figure 20: Why a "best fit" for g is a statement about the prior. The likelihood depends on the product $g\alpha$, so the misfit surface is a valley running along constant $g\alpha$ and the data is flat along g/α. An optimiser slides until it meets the edge of the box, and *where* it stops is set by the box: the two dashed rectangles give the two filled points, which differ in g by more than thirtyfold. This is the mechanism behind the fifty-nat swing in the operator ranking of Section 5 when the bounds were widened, and behind the order-of-magnitude disagreement in g between methods in Figure 8.

terms, i.e. using $\mathrm{diag}(C)$, does not merely lose efficiency – it misstates the amount of information. For a stationary sequence with autocorrelation ρ_k,

$$\mathrm{Var}\left(\sum_{i=1}^{N} r_i\right) = N\sigma^2 \left(1 + 2\sum_{k=1}^{N-1}\left(1 - \tfrac{k}{N}\right)\rho_k\right) \xrightarrow{N\to\infty} N\sigma^2 \left(1 + 2\sum_{k\geq 1}\rho_k\right), \tag{47}$$

so the sequence carries the information of $N_{\text{eff}} = N/(1 + 2\sum \rho_k)$ independent draws. The bracket is the spectral density of the residual sequence at zero frequency, which is the cleaner way to see why a long-range correlation is so costly.

Correlated residuals in a *fitted* model usually indicate model discrepancy rather than correlated measurement noise. Kennedy and O'Hagan [12] formalise this by writing $y = m(\theta) + \delta(\cdot) + \varepsilon$ with δ a Gaussian process, and Brynjarsdóttir and O'Hagan show that omitting δ leaves parameter estimates biased *and* overconfident, while an unconstrained δ absorbs the physics. Neither failure is visible in χ^2.

A.9 Optimal experimental design

A design ξ determines what is measured and hence $F(\xi, \theta)$. Since F is a matrix and designs must be ranked, a scalar functional is required, and the classical choices are

$$\Phi_D = \log\det F, \quad \Phi_A = -\operatorname{tr} F^{-1}, \quad \Phi_E = \lambda_{\min}(F), \quad \Phi_c = -c^{\mathsf{T}} F^{-1} c, \tag{48}$$

each maximised. In Bayesian terms, Φ_D is the expected Kullback–Leibler gain from prior to posterior under a Gaussian approximation, which is why it is the default: it maximises information about *all* parameters jointly. That default is wrong whenever one direction is the question, because $\log\det F = \sum_k \log\lambda_k$ is dominated by the large eigenvalues.

Because F depends on θ, a local design presupposes an answer. The standard repairs are to average over a prior, $\Phi(\xi) = \mathbb{E}_\theta[\Phi(F(\xi,\theta))]$, or to take the worst case over a plausible set. Sequential design uses the posterior from completed experiments as the prior for the next, which is what is done here.

When the model itself is uncertain, all of the above presuppose the wrong thing. T-optimality (Atkinson and Fedorov) maximises instead the lack of fit of the inferior model,

$$\Phi_T(\xi) = \min_{\theta_2} \left\| m_1(\hat{\theta}_1; \xi) - m_2(\theta_2; \xi) \right\|^2, \tag{49}$$

which asks which experiment the candidates most disagree about. Its virtue here is that it shares no assumptions with the Fisher criteria, so agreement between them is evidence rather than restatement.

B Reproducing

```
.venv/bin/python examples/measured_selection.py <dataset> [thermal Nt] [workers]
.venv/bin/python docs/writeup/collect.py 16     # records results.json
.venv/bin/python docs/writeup/figures.py        # draws every figure from it
```

Every number and figure in this document comes from `results.json`; none is transcribed by hand.

Code and data availability

Every number and figure in this document is reproduced by scripts in the repository that contains it: `figures.py`, `figures_geometry.py`, `figures_identifiability.py`, `figures_measure.py` and `figures_trial.py` draw the figures, `fit_quality.py` the fit table and `trial_variation.py` the trial-spread analysis, each from the tracked `results.json`. The design measures use `pyimr.measure`, the discrimination criteria `pyimr.discriminate`, and the trade-off front `pyimr.pareto`. The measured records themselves are not redistributed here.

References

[1] Joseph B. Keller and Michael Miksis. Bubble oscillations of large amplitude. *The Journal of the Acoustical Society of America*, 68(2):628–633, 1980.

[2] Jonathan B. Freund and Randy H. Ewoldt. Quantitative rheological model selection: Good fits versus credible models using Bayesian inference. *Journal of Rheology*, 59(3):667–701, 2015.

[3] Charles J. Geyer. Practical Markov chain Monte Carlo. *Statistical Science*, 7(4):473–483, 1992.

[4] A. C. Atkinson and V. V. Fedorov. The design of experiments for discriminating between two rival models. *Biometrika*, 62(1):57–70, 1975.

[5] D. V. Lindley. A statistical paradox. *Biometrika*, 44(1–2):187–192, 1957.

[6] Harold Jeffreys. Some tests of significance, treated by the theory of probability. *Mathematical Proceedings of the Cambridge Philosophical Society*, 31(2):203–222, 1935.

[7] J. Kiefer and J. Wolfowitz. The equivalence of two extremum problems. *Canadian Journal of Mathematics*, 12:363–366, 1960.

[8] S. D. Silvey, D. M. Titterington, and B. Torsney. An algorithm for optimal designs on a finite design space. *Communications in Statistics — Theory and Methods*, A7:1379–1389, 1978.

[9] A. C. Atkinson. DT-optimum designs for model discrimination and parameter estimation. *Journal of Statistical Planning and Inference*, 138(1):56–64, 2008.

[10] R. D. Cook and W. K. Wong. On the equivalence of constrained and compound optimal designs. *Journal of the American Statistical Association*, 89(426):687–692, 1994.

[11] Ryan N. Gutenkunst, Joshua J. Waterfall, Fergal P. Casey, Kevin S. Brown, Christopher R. Myers, and James P. Sethna. Universally sloppy parameter sensitivities in systems biology models. *PLoS Computational Biology*, 3(10):e189, 2007.

[12] Marc C. Kennedy and Anthony O'Hagan. Bayesian calibration of computer models. *Journal of the Royal Statistical Society: Series B*, 63(3):425–464, 2001.